Detecting AI-Generated vs Human-Written Text

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How to Run

1. Open this notebook in Google Colab.
2. Upload `AI_Human.csv.zip` to the Colab environment.
3. Run all cells from top to bottom.
4. The notebook will:
    - Load the dataset
    - Train baseline models
    - Train machine learning/deep learning models
    - Evaluate models on validation and test sets
    - Run external validation
    - Save final results to CSV

Dataset

The main dataset file required is:

`AI_Human.csv.zip`

Link: https://www.kaggle.com/datasets/shanegerami/ai-vs-human-text

Expected columns:

- `text`
- `generated`

Label meaning:

- `0 = Human-written`
- `1 = AI-generated`

```
# Install required packages
!pip install transformers datasets accelerate sentencepiece -q
!pip install wordcloud -q

import os
os.environ["TF_CPP_MIN_LOG_LEVEL"] = "2"
os.environ["TOKENIZERS_PARALLELISM"] = "false"
print("Libraries installed")
```

```
Libraries installed
```

```
# Import all libraries
# ── Standard ──────────────────────────────────────────────
import re, random, warnings, zipfile
import numpy as np
import pandas as pd

# ── Visualization ─────────────────────────────────────────
import matplotlib.pyplot as plt
import seaborn as sns
from wordcloud import WordCloud

# ── Scikit-learn ──────────────────────────────────────────
from sklearn.model_selection import train_test_split
from sklearn.dummy import DummyClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.feature_extraction.text import TfidfVectorizer, ENGLISH_S
from sklearn.metrics import (
    accuracy_score, precision_score, recall_score,
    f1_score, classification_report, confusion_matrix, ConfusionMatri
)

# ── TensorFlow / Keras (LSTM trained from scratch) ────────
import tensorflow as tf
from tensorflow.keras.preprocessing.text import Tokenizer
from tensorflow.keras.preprocessing.sequence import pad_sequences
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import (
    Embedding, LSTM, Dense, Dropout,
    Conv1D, GlobalMaxPooling1D, Bidirectional
)
from tensorflow.keras.callbacks import EarlyStopping

# ── PyTorch + HuggingFace Transformers ────────────────────
```

```
import torch
from torch.utils.data import Dataset, DataLoader
from transformers import (
    AutoTokenizer, AutoModelForSequenceClassification,
    AutoModel, get_linear_schedule_with_warmup
)
from torch.optim import AdamW

warnings.filterwarnings("ignore")

# ── GPU Check ──────────────────────────────────────────────
print("TensorFlow GPU:", tf.config.list_physical_devices("GPU"))
print("PyTorch CUDA:", torch.cuda.is_available())
if torch.cuda.is_available():
    print("GPU:", torch.cuda.get_device_name(0))

DEVICE = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print("Using device:", DEVICE)
```

```
TensorFlow GPU: [PhysicalDevice(name='/physical_device:GPU:0', device_t
PyTorch CUDA: True
GPU: Tesla T4
Using device: cuda
```

Set Random Seeds

```
SEED = 42
random.seed(SEED)
np.random.seed(SEED)
tf.random.set_seed(SEED)
torch.manual_seed(SEED)
if torch.cuda.is_available():
    torch.cuda.manual_seed_all(SEED)
print("Seeds set to", SEED)
```

```
Seeds set to 42
```

Load Dataset

```
# ── Load from uploaded ZIP ──────────────────────
zip_path    = "/content/AI_Human.csv.zip"
extract_dir = "/content/dataset"
os.makedirs(extract_dir, exist_ok=True)

with zipfile.ZipFile(zip_path, 'r') as z:
    z.extractall(extract_dir)

csv_path = os.path.join(extract_dir, "AI_Human.csv")
df = pd.read_csv(csv_path)

# ── Sanity check ────────────────────────────────────────
TEXT_COL  = "text"
LABEL_COL = "generated"

df[LABEL_COL] = df[LABEL_COL].astype(int)
df = df.drop_duplicates(subset=[TEXT_COL]).reset_index(drop=True)

print("Shape:", df.shape)
print(df[LABEL_COL].value_counts())
display(df.head(3))
```

```
Shape: (487235, 2)
generated
0    305797
1    181438
Name: count, dtype: int64
```

	text	generated
0	Cars. Cars have been around since they became ...	0
1	Transportation is a large necessity in most co...	0
2	"America's love affair with it's vehicles seem...	0

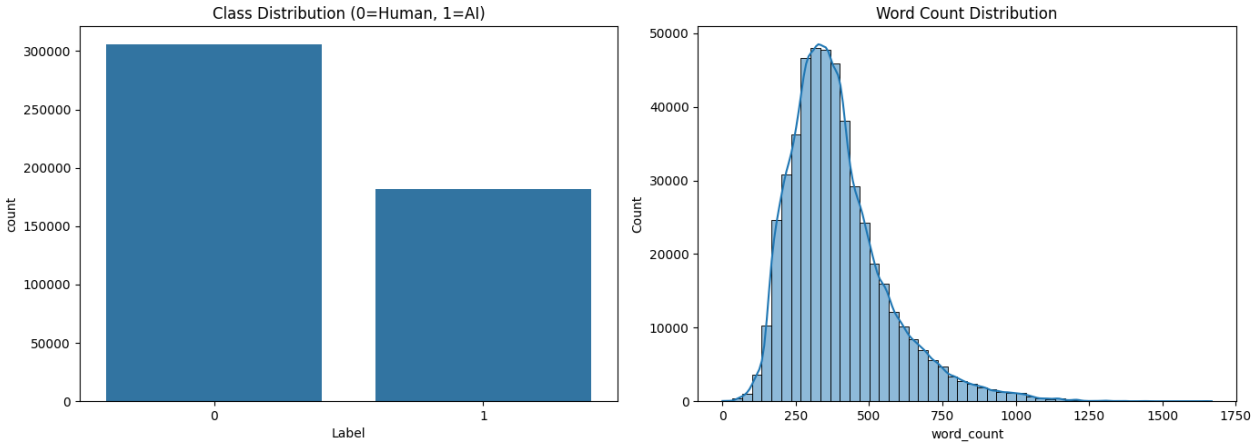
Exploratory Data Analysis (EDA)

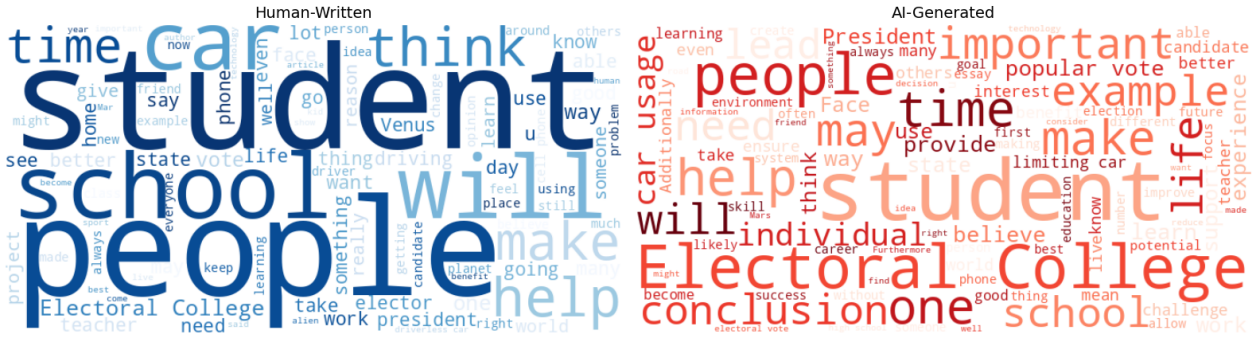
```
# ── Text length features ────────────────────────────────
df["word_count"]     = df[TEXT_COL].apply(lambda x: len(x.split()))
df["char_count"]     = df[TEXT_COL].apply(len)
df["sent_count"]     = df[TEXT_COL].apply(
    lambda x: len([s for s in re.split(r"[.!?]+", x) if s.strip()])
)
```

```
df["avg_word_len"]   = df[TEXT_COL].apply(
    lambda x: np.mean([len(w) for w in x.split()]) if x.split() else (
)
df["punct_count"]    = df[TEXT_COL].apply(
    lambda x: len(re.findall(r"[.,!?;:]", x))
)

stop_words = set(ENGLISH_STOP_WORDS)
df["stopword_ratio"] = df[TEXT_COL].apply(
    lambda x: sum(1 for w in x.lower().split() if w in stop_words) / r
)

print(df[["word_count","char_count","sent_count","avg_word_len"]].desc

# ── Plots ──────────────────────────────────────────────────
fig, axes = plt.subplots(2, 2, figsize=(14, 10))

# Class distribution
sns.countplot(x=LABEL_COL, data=df, ax=axes[0,0])
axes[0,0].set_title("Class Distribution (0=Human, 1=AI)")
axes[0,0].set_xlabel("Label")

# Word count distribution
sns.histplot(df["word_count"], bins=50, kde=True, ax=axes[0,1])
axes[0,1].set_title("Word Count Distribution")

# Word count by class
sns.boxplot(x=LABEL_COL, y="word_count", data=df, ax=axes[1,0])
axes[1,0].set_title("Word Count by Class")

# Avg word length by class
sns.boxplot(x=LABEL_COL, y="avg_word_len", data=df, ax=axes[1,1])
axes[1,1].set_title("Avg Word Length by Class")

plt.tight_layout()
plt.show()

# ── Word clouds ────────────────────────────────────────────
fig, axes = plt.subplots(1, 2, figsize=(16, 5))
for label, title, ax in zip([0, 1], ["Human-Written", "AI-Generated"],
    text = " ".join(df[df[LABEL_COL]==label][TEXT_COL].sample(500, ran
    wc   = WordCloud(width=700, height=350, background_color="white",
                     max_words=100, colormap="Blues" if label==0 else
    ax.imshow(wc, interpolation="bilinear")
    ax.set_title(title, fontsize=14)
```

```
    ax.axis("off")
plt.tight_layout()
plt.show()
```

	word_count	char_count	sent_count	avg_word_len
count	487235.00	487235.00	487235.00	487235.00
mean	393.10	2269.59	20.66	4.76
std	168.59	988.81	9.20	0.52
min	0.00	1.00	0.00	0.00
25%	278.00	1583.00	14.00	4.42
50%	363.00	2102.00	20.00	4.69
75%	471.00	2724.00	26.00	5.02
max	1668.00	18322.00	155.00	126.00





Text Cleaning

```
def clean_text(text: str) -> str:
    """
    Lightweight cleaning used for classical ML + LSTM.
    NOTE: BERT/Qwen use RAW text — transformers handle their own toke
    """
    text = text.lower()
    text = re.sub(r"http\S+|www\S+", "", text)   # remove URLs
    text = re.sub(r"<.*?>", "", text)              # remove HTML
    text = re.sub(r"[^a-zA-Z\s]", " ", text)       # keep letters onl
    text = re.sub(r"\s+", " ", text).strip()
    return text

df["clean_text"] = df[TEXT_COL].apply(clean_text)
display(df[[TEXT_COL, "clean_text"]].head(3))
print("Text cleaning complete")
```

	text	clean_text
0	Cars. Cars have been around since they became ...	cars cars have been around since they became f...
1	Transportation is a large necessity in most co...	transportation is a large necessity in most co...
2	"America's love affair with it's vehicles	america s love affair with it s vehicles

Train / Validation / Test Split (70 / 15 / 15)

```
X_clean = df["clean_text"]
X_raw   = df[TEXT_COL]          # raw text — used by BERT / Qwen
y       = df[LABEL_COL]

# First split: 70% train, 30% temp
Xc_tr, Xc_tmp, Xr_tr, Xr_tmp, y_tr, y_tmp = train_test_split(
    X_clean, X_raw, y, test_size=0.30, random_state=SEED, stratify=y
)

# Second split: 15% val, 15% test (from 30% temp)
Xc_val, Xc_test, Xr_val, Xr_test, y_val, y_test = train_test_split(
    Xc_tmp, Xr_tmp, y_tmp, test_size=0.50, random_state=SEED, stratif
)

print(f"Train : {len(Xc_tr):>7,}")
print(f"Val   : {len(Xc_val):>7,}")
print(f"Test  : {len(Xc_test):>7,}")
```

```
Train : 341,064
Val   :  73,085
Test  :  73,086
```

Evaluation Helper

```
results = []   # stores all model results for final comparison

def evaluate_model(y_true, y_pred, model_name: str, split="Validation'
    """Compute metrics, print report, draw confusion matrix."""
    acc  = accuracy_score(y_true, y_pred)
    prec = precision_score(y_true, y_pred, zero_division=0)
    rec  = recall_score(y_true, y_pred, zero_division=0)
    f1   = f1_score(y_true, y_pred, zero_division=0)

    print(f"\n{'='*50}")
    print(f" {model_name} [{split}]")
    print(f"{'='*50}")
    print(f"  Accuracy : {acc:.4f}")
    print(f"  Precision: {prec:.4f}")
    print(f"  Recall   : {rec:.4f}")
    print(f"  F1-score : {f1:.4f}")
    print()
    print(classification_report(y_true, y_pred, target_names=["Human",

    cm   = confusion_matrix(y_true, y_pred)
    disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=
    disp.plot(cmap="Blues")
    plt.title(f"Confusion Matrix — {model_name}")
    plt.show()

    return {"Model": model_name, "Split": split,
            "Accuracy": acc, "Precision": prec, "Recall": rec, "F1": f

print("✅ evaluate_model() ready")
```

```
✅ evaluate_model() ready
```

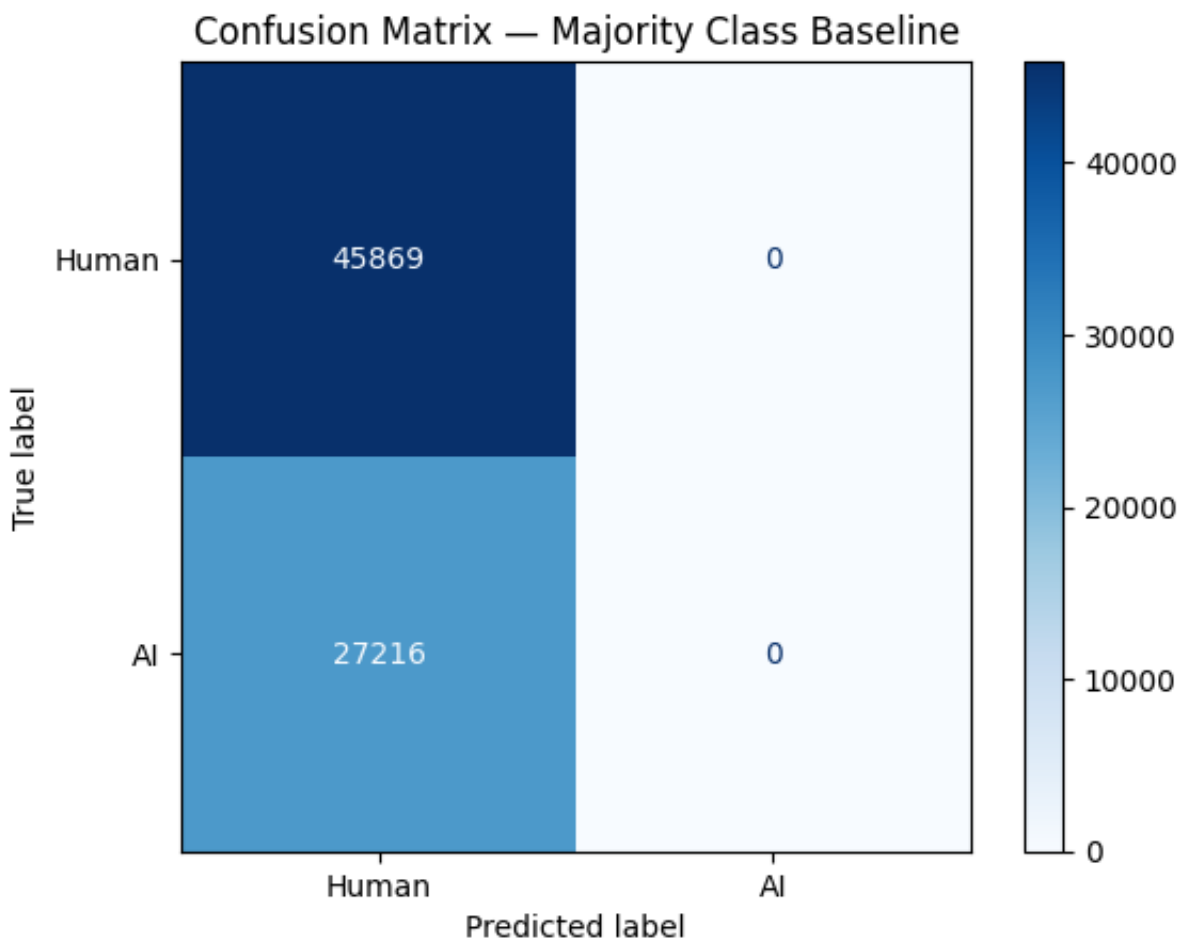
Baseline Models

(Majority class & Stratified — reference points only)

```
for strategy, name in [("most_frequent", "Majority Class Baseline"),
                       ("stratified",    "Stratified Baseline")]:
    clf = DummyClassifier(strategy=strategy, random_state=SEED)
    clf.fit(Xc_tr.to_frame(), y_tr)
    y_pred = clf.predict(Xc_val.to_frame())
```

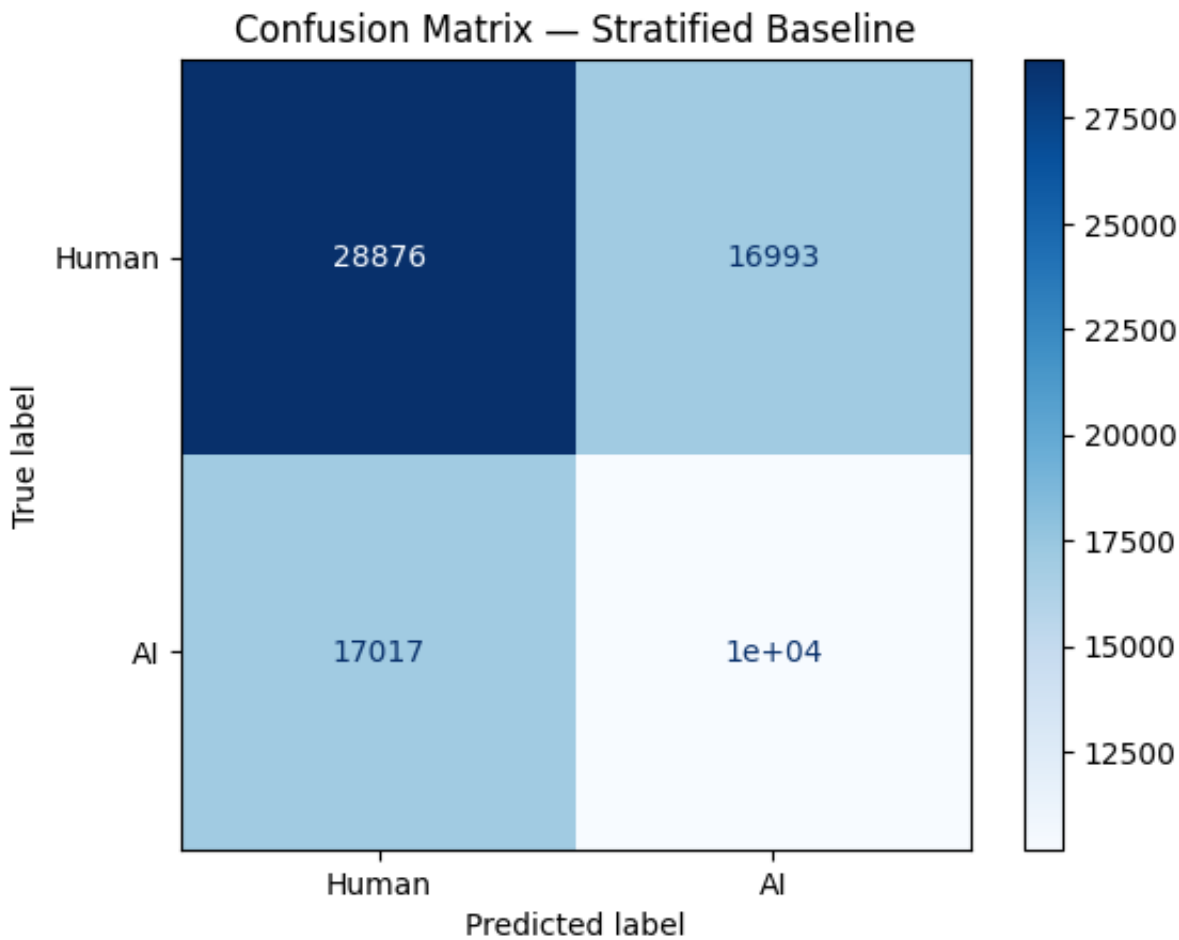
```
    results.append(evaluate_model(y_val, y_pred, name))
```

```
==================================================
 Majority Class Baseline [Validation]
==================================================
  Accuracy : 0.6276
  Precision: 0.0000
  Recall   : 0.0000
  F1-score : 0.0000

              precision    recall  f1-score   support

       Human       0.63      1.00      0.77     45869
          AI       0.00      0.00      0.00     27216

    accuracy                           0.63     73085
   macro avg       0.31      0.50      0.39     73085
weighted avg       0.39      0.63      0.48     73085
```




```
==================================================
 Stratified Baseline [Validation]
--------------------------------------------------
```

```
--------------------------------------------------
  Accuracy : 0.5347
  Precision: 0.3751
  Recall   : 0.3747
  F1-score : 0.3749

              precision    recall  f1-score   support

       Human       0.63      0.63      0.63     45869
          AI       0.38      0.37      0.37     27216

    accuracy                           0.53     73085
   macro avg       0.50      0.50      0.50     73085
weighted avg       0.53      0.53      0.53     73085
```

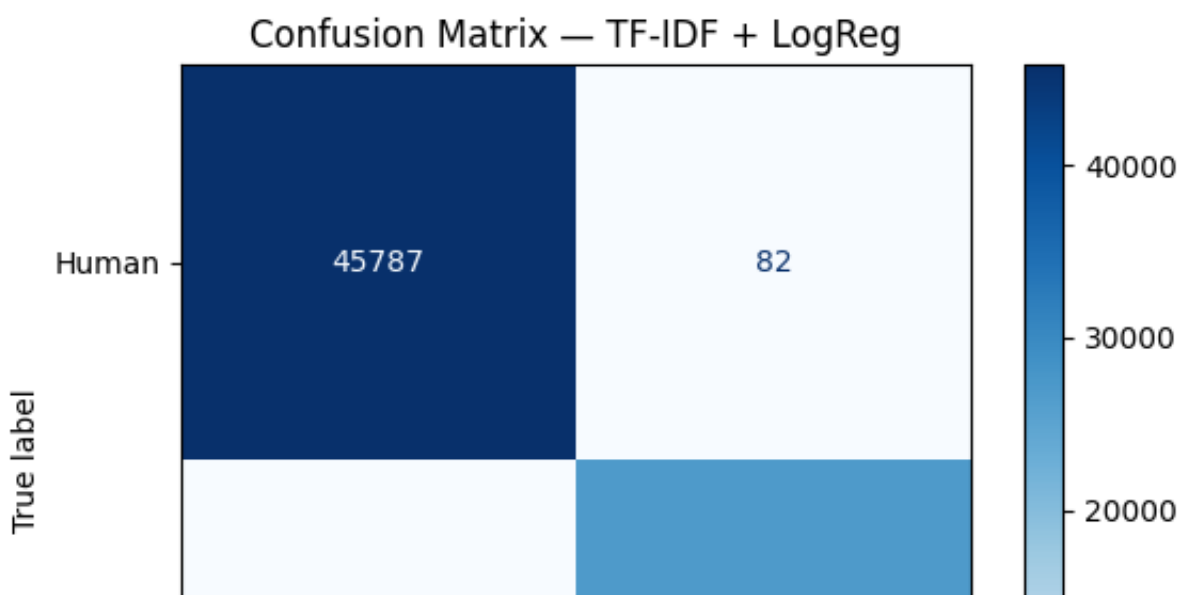


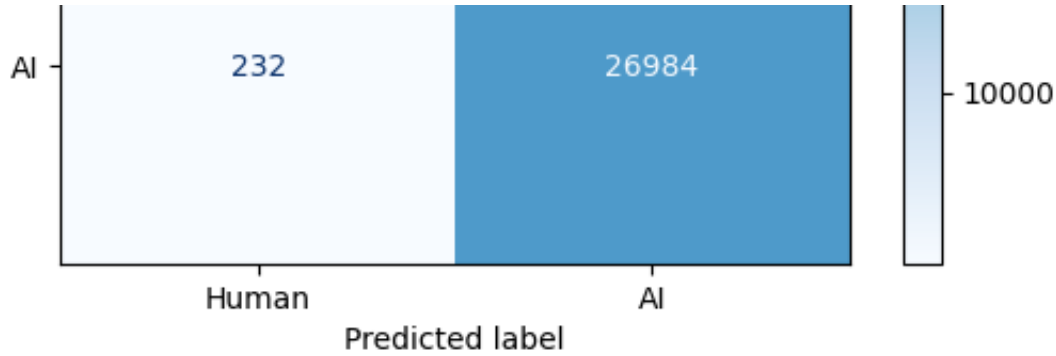
TF-IDF + Logistic Regression

(Sparse bag-of-words — no embedding)

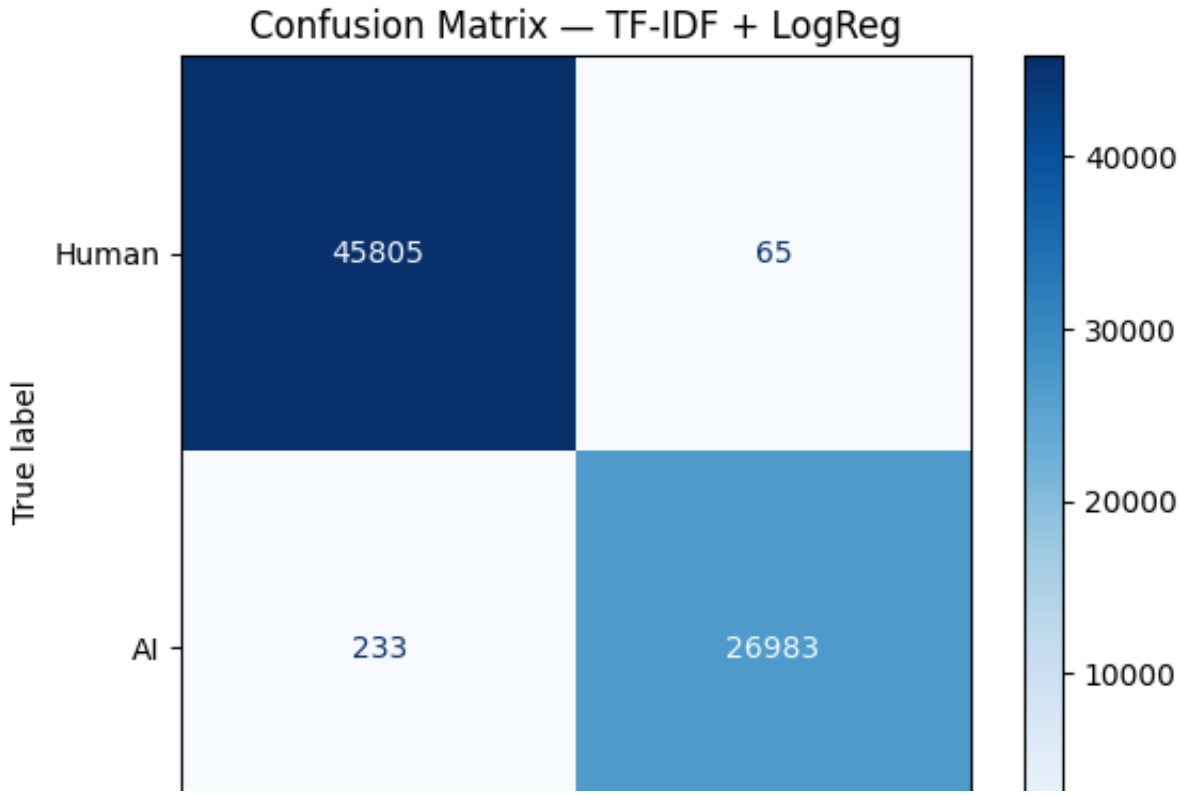
```
# ── Vectorise ──────────────────────────────────────────────
tfidf = TfidfVectorizer(max_features=10_000, ngram_range=(1,2), min_d
X_tr_tfidf  = tfidf.fit_transform(Xc_tr)
X_val_tfidf = tfidf.transform(Xc_val)
X_te_tfidf  = tfidf.transform(Xc_test)

# ── Train ──────────────────────────────────────────────────
lr = LogisticRegression(max_iter=1000, random_state=SEED, C=1.0)
lr.fit(X_tr_tfidf, y_tr)

# ── Val + Test ─────────────────────────────────────────────
results.append(evaluate_model(y_val,  lr.predict(X_val_tfidf), "TF-IDI
results.append(evaluate_model(y_test, lr.predict(X_te_tfidf),  "TF-IDI
```

```
==================================================
 TF-IDF + LogReg [Validation]
==================================================
  Accuracy : 0.9957
  Precision: 0.9970
  Recall   : 0.9915
  F1-score : 0.9942

              precision    recall  f1-score   support

       Human       0.99      1.00      1.00     45869
          AI       1.00      0.99      0.99     27216

    accuracy                           1.00     73085
   macro avg       1.00      0.99      1.00     73085
weighted avg       1.00      1.00      1.00     73085
```






```
==================================================
 TF-IDF + LogReg [Test]
==================================================
  Accuracy : 0.9959
  Precision: 0.9976
  Recall   : 0.9914
  F1-score : 0.9945

              precision    recall  f1-score   support

       Human       0.99      1.00      1.00     45870
          AI       1.00      0.99      0.99     27216

    accuracy                           1.00     73086
   macro avg       1.00      1.00      1.00     73086
weighted avg       1.00      1.00      1.00     73086
```



LSTM with Embeddings Trained FROM SCRATCH

```
MAX_VOCAB = 20_000
MAX_LEN   = 256    # increased from 200 for better context capture
EMBED_DIM = 128    # embedding dimension trained from scratch

# ── Tokenize ──────────────────────────────────────────────
tok = Tokenizer(num_words=MAX_VOCAB, oov_token="<OOV>")
tok.fit_on_texts(Xc_tr)                                # fit ONLY on t

def to_padded(texts):
    seqs = tok.texts_to_sequences(texts)
    return pad_sequences(seqs, maxlen=MAX_LEN, padding="post", trunca

X_tr_pad  = to_padded(Xc_tr)
X_val_pad = to_padded(Xc_val)
X_te_pad  = to_padded(Xc_test)

print(f"Train padded: {X_tr_pad.shape}")
```

```
Train padded: (341064, 256)
```

```
# ── Architecture ──────────────────────────────────────────
# Embedding layer is randomly initialised and TRAINED FROM SCRATCH
lstm_model = Sequential([
    Embedding(input_dim=MAX_VOCAB,     # vocabulary size
              output_dim=EMBED_DIM,    # 128-dim dense vectors (learn
              input_length=MAX_LEN,
              name="scratch_embedding"),   # <-- trained from scratch
    Bidirectional(LSTM(64, return_sequences=True)),
    Dropout(0.3),
    Bidirectional(LSTM(32)),
    Dropout(0.3),
    Dense(64, activation="relu"),
    Dropout(0.3),
    Dense(1, activation="sigmoid")
```

```
], name="BiLSTM_scratch_embedding")

lstm_model.compile(optimizer="adam",
                   loss="binary_crossentropy",
                   metrics=["accuracy"])
lstm_model.summary()

print("\n📌 Embedding layer is randomly initialised and trained from
```

Model: "BiLSTM_scratch_embedding"

Layer (type)	Output Shape	Para
scratch_embedding (Embedding)	?	0 (unbu
bidirectional (Bidirectional)	?	0 (unbu
dropout (Dropout)	?	
bidirectional_1 (Bidirectional)	?	0 (unbu
dropout_1 (Dropout)	?	
dense (Dense)	?	0 (unbu
dropout_2 (Dropout)	?	
dense_1 (Dense)	?	0 (unbu

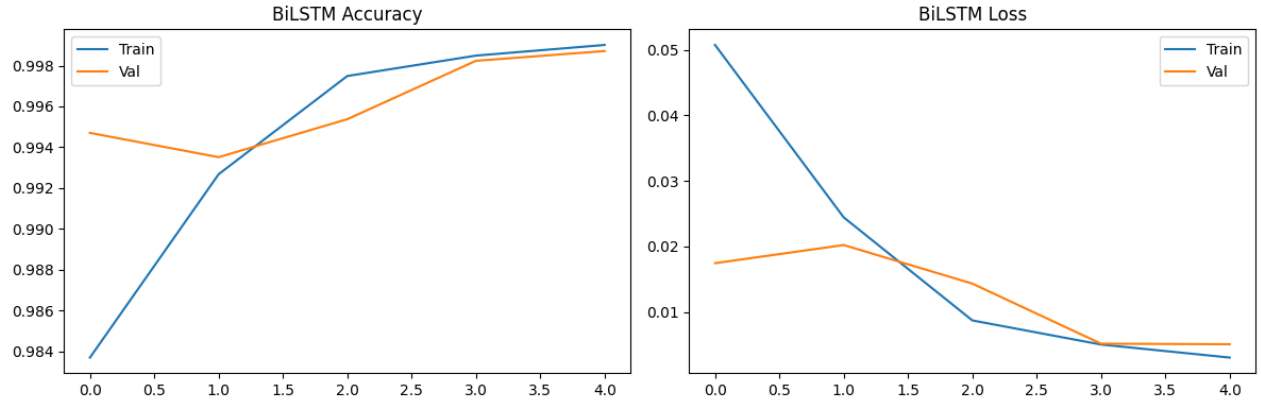
Total params: 0 (0.00 B)
Trainable params: 0 (0.00 B)
Non-trainable params: 0 (0.00 B)

📌 Embedding layer is randomly initialised and trained from scratch

```
# ── Train ──────────────────────────────────────────────────
early_stop = EarlyStopping(monitor="val_loss", patience=2, restore_bes

history_lstm = lstm_model.fit(
    X_tr_pad, y_tr,
    validation_data=(X_val_pad, y_val),
    epochs=5,
    batch_size=64,
    callbacks=[early_stop]
)

# ── Learning curves ────────────────────────────────────────
fig, axes = plt.subplots(1, 2, figsize=(12, 4))
```

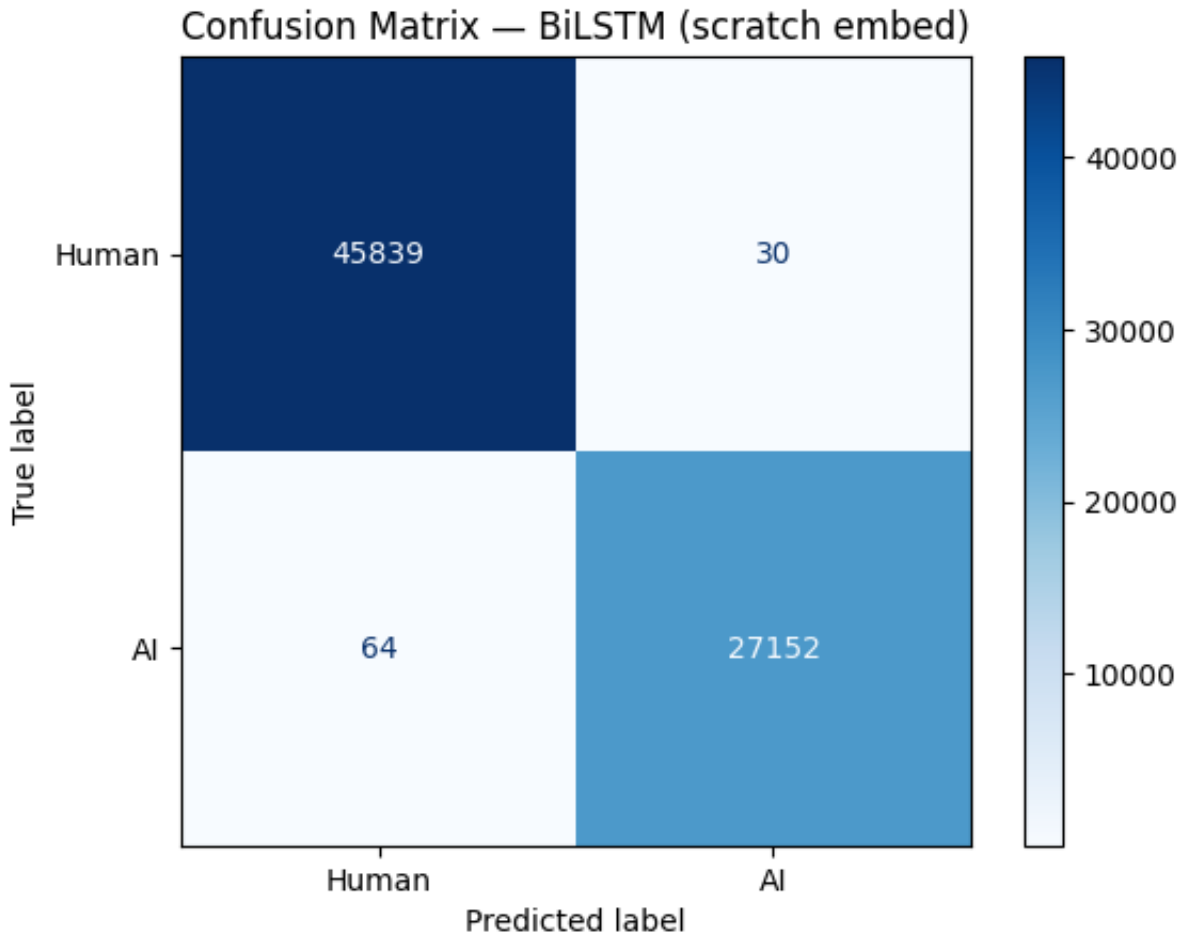
```
for ax, metric, title in zip(axes, ["accuracy","loss"], ["Accuracy","l
    ax.plot(history_lstm.history[metric],     label="Train")
    ax.plot(history_lstm.history[f"val_{metric}"], label="Val")
    ax.set_title(f"BiLSTM {title}")
    ax.legend()
plt.tight_layout(); plt.show()

# ── Evaluate ──────────────────────────────────────────────
y_pred_lstm_val  = (lstm_model.predict(X_val_pad)  > 0.5).astype(int)
y_pred_lstm_test = (lstm_model.predict(X_te_pad)   > 0.5).astype(int)
results.append(evaluate_model(y_val,  y_pred_lstm_val,  "BiLSTM (scra
results.append(evaluate_model(y_test, y_pred_lstm_test, "BiLSTM (scra
```

```
Epoch 1/5
5330/5330 ━━━━━━━━━━━━━━━━━━━━ 238s 43ms/step - accuracy: 0.9837 - loss
Epoch 2/5
5330/5330 ━━━━━━━━━━━━━━━━━━━━ 228s 43ms/step - accuracy: 0.9927 - loss
Epoch 3/5
5330/5330 ━━━━━━━━━━━━━━━━━━━━ 230s 43ms/step - accuracy: 0.9975 - loss
Epoch 4/5
5330/5330 ━━━━━━━━━━━━━━━━━━━━ 229s 43ms/step - accuracy: 0.9985 - loss
Epoch 5/5
5330/5330 ━━━━━━━━━━━━━━━━━━━━ 230s 43ms/step - accuracy: 0.9990 - loss
```

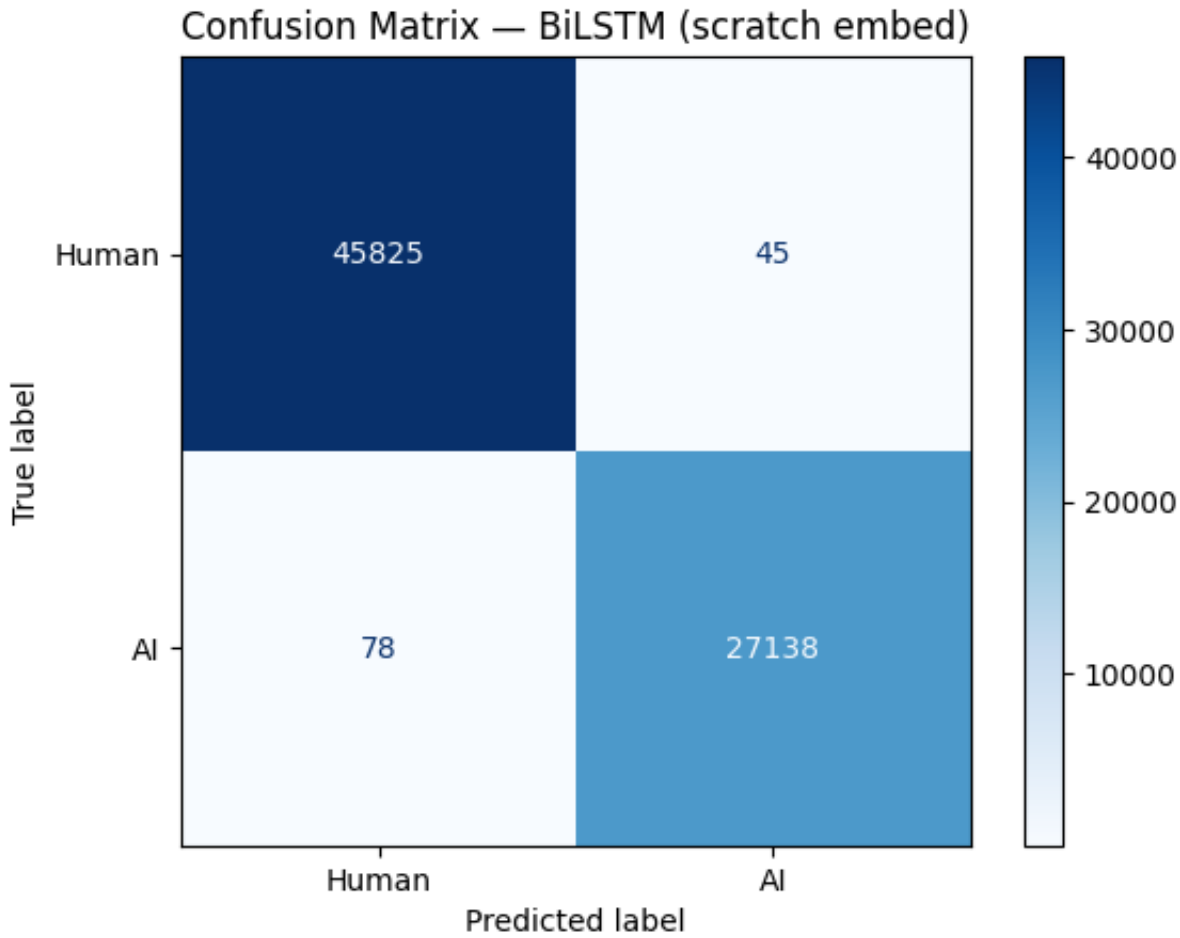



```
2284/2284 ━━━━━━━━━━━━━━━━━━━━ 36s 16ms/step
2284/2284 ━━━━━━━━━━━━━━━━━━━━ 35s 15ms/step

==================================================
 BiLSTM (scratch embed) [Validation]
==================================================
  Accuracy : 0.9987
  Precision: 0.9989
  Recall   : 0.9976
  F1-score : 0.9983

              precision    recall  f1-score   support

       Human       1.00      1.00      1.00     45869
          AI       1.00      1.00      1.00     27216
```

```
          AI       1.00      1.00      1.00     27216

    accuracy                           1.00     73085
   macro avg       1.00      1.00      1.00     73085
weighted avg       1.00      1.00      1.00     73085
```




```
==================================================
 BiLSTM (scratch embed) [Test]
==================================================
  Accuracy : 0.9983
  Precision: 0.9983
  Recall   : 0.9971
  F1-score : 0.9977

              precision    recall  f1-score   support

       Human       1.00      1.00      1.00     45870
          AI       1.00      1.00      1.00     27216

    accuracy                           1.00     73086
   macro avg       1.00      1.00      1.00     73086
weighted avg       1.00      1.00      1.00     73086
```



BERT Fine-Tuning

(Pretrained contextual embeddings — fine-tuned on our task)

```
# ── PyTorch Dataset wrapper ──────────────────────────────
class TextDataset(Dataset):
    def __init__(self, texts, labels, tokenizer, max_len=128):
        self.texts     = list(texts)
        self.labels    = list(labels)
```

```
        self.tokenizer = tokenizer
        self.max_len   = max_len

    def __len__(self):
        return len(self.texts)

    def __getitem__(self, idx):
        enc = self.tokenizer(
            self.texts[idx],
            max_length=self.max_len,
            padding="max_length",
            truncation=True,
            return_tensors="pt"
        )
        return {
            "input_ids":      enc["input_ids"].squeeze(0),
            "attention_mask": enc["attention_mask"].squeeze(0),
            "label":          torch.tensor(self.labels[idx], dtype=to
        }


def train_transformer(model_name: str,
                      X_tr, X_val, X_te,
                      y_tr, y_val, y_te,
                      max_len=128, batch_size=16,
                      epochs=2, lr=2e-5):
    """
    Fine-tune any HuggingFace classification model.
    Returns val predictions, test predictions.
    """
    print(f"\n🔄 Loading {model_name} ...")
    tokenizer = AutoTokenizer.from_pretrained(model_name)
    model     = AutoModelForSequenceClassification.from_pretrained(
                    model_name, num_labels=2).to(DEVICE)

    # Datasets + loaders
    train_ds = TextDataset(X_tr,  y_tr,  tokenizer, max_len)
    val_ds   = TextDataset(X_val, y_val, tokenizer, max_len)
    test_ds  = TextDataset(X_te,  y_te,  tokenizer, max_len)

    train_loader = DataLoader(train_ds, batch_size=batch_size, shuffl
    val_loader   = DataLoader(val_ds,   batch_size=batch_size*2)
    test_loader  = DataLoader(test_ds,  batch_size=batch_size*2)

    # Optimiser + scheduler
```

```
optimizer = AdamW(model.parameters(), lr=lr, weight_decay=0.01)
total_steps = len(train_loader) * epochs
scheduler = get_linear_schedule_with_warmup(
               optimizer,
               num_warmup_steps=total_steps // 10,
               num_training_steps=total_steps)

train_accs, val_accs = [], []

for epoch in range(1, epochs+1):
    # ── Train ──────────────────────────────────────────
    model.train()
    total_loss, correct, total = 0, 0, 0
    for batch in train_loader:
        ids  = batch["input_ids"].to(DEVICE)
        mask = batch["attention_mask"].to(DEVICE)
        lbls = batch["label"].to(DEVICE)

        optimizer.zero_grad()
        out  = model(input_ids=ids, attention_mask=mask, labels=lI
        loss = out.loss
        loss.backward()
        torch.nn.utils.clip_grad_norm_(model.parameters(), 1.0)
        optimizer.step()
        scheduler.step()

        total_loss += loss.item()
        preds = out.logits.argmax(-1)
        correct += (preds == lbls).sum().item()
        total   += len(lbls)

    tr_acc = correct / total
    train_accs.append(tr_acc)

    # ── Validate ───────────────────────────────────────
    model.eval()
    v_preds, v_labels = [], []
    with torch.no_grad():
        for batch in val_loader:
            ids  = batch["input_ids"].to(DEVICE)
            mask = batch["attention_mask"].to(DEVICE)
            out  = model(input_ids=ids, attention_mask=mask)
            v_preds.extend(out.logits.argmax(-1).cpu().numpy())
            v_labels.extend(batch["label"].numpy())
```

```
            val_acc = accuracy_score(v_labels, v_preds)
            val_accs.append(val_acc)
            print(f"  Epoch {epoch}/{epochs} | Train Acc: {tr_acc:.4f} | V

        # ── Learning curves ──────────────────────────────────
        plt.figure(figsize=(7,4))
        plt.plot(train_accs, label="Train", marker="o")
        plt.plot(val_accs,   label="Val",   marker="o")
        plt.title(f"{model_name} — Accuracy per Epoch")
        plt.legend(); plt.show()

        # ── Test predictions ─────────────────────────────────
        model.eval()
        t_preds = []
        with torch.no_grad():
            for batch in test_loader:
                ids  = batch["input_ids"].to(DEVICE)
                mask = batch["attention_mask"].to(DEVICE)
                out  = model(input_ids=ids, attention_mask=mask)
                t_preds.extend(out.logits.argmax(-1).cpu().numpy())

        return np.array(v_preds), np.array(t_preds), model, tokenizer

    print("✅ train_transformer() defined")
```

```
✅ train_transformer() defined
```

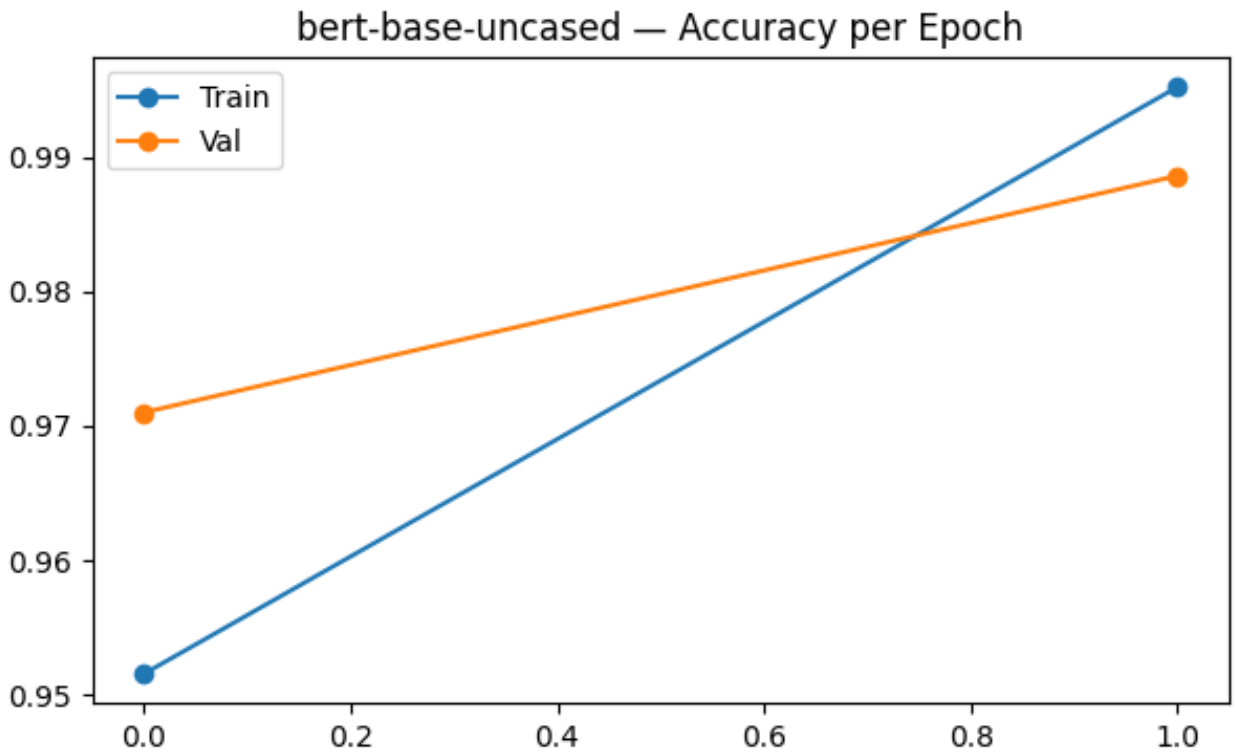
```
# ── NOTE on subsetting ──────────────────────────────────────
# Fine-tuning on 487k samples takes ~hours per epoch on a T4.
# We subsample for feasibility; increase N_FINETUNE for a final run.
N_FINETUNE = 20_000   # ← set to None to use full dataset

if N_FINETUNE:
    idx_tr  = np.random.choice(len(Xr_tr),  N_FINETUNE, replace=False
    idx_val = np.random.choice(len(Xr_val), N_FINETUNE//4, replace=Fa
    idx_te  = np.random.choice(len(Xr_test),N_FINETUNE//4, replace=Fa
    Xr_tr_s  = Xr_tr.iloc[idx_tr].reset_index(drop=True)
    Xr_val_s = Xr_val.iloc[idx_val].reset_index(drop=True)
    Xr_te_s  = Xr_test.iloc[idx_te].reset_index(drop=True)
    y_tr_s   = y_tr.iloc[idx_tr].reset_index(drop=True)
    y_val_s  = y_val.iloc[idx_val].reset_index(drop=True)
    y_te_s   = y_test.iloc[idx_te].reset_index(drop=True)
    print(f"Using subset: train={len(Xr_tr_s)}, val={len(Xr_val_s)}, 
else:
    Xr_tr_s, Xr_val_s, Xr_te_s = Xr_tr, Xr_val, Xr_test
    y_tr_s,  y_val_s,  y_te_s  = y_tr,  y_val,  y_test
```

```
Using subset: train=20000, val=5000, test=5000
```

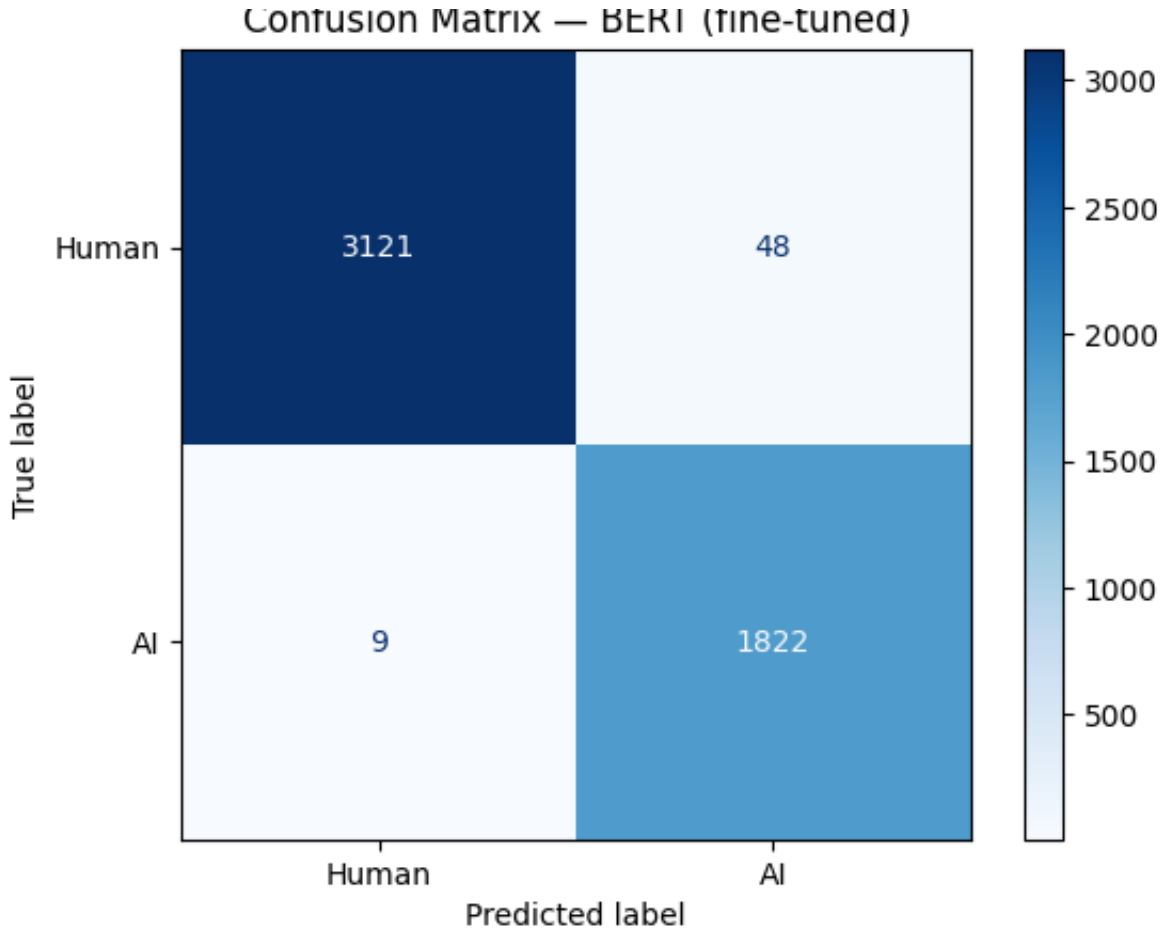
```
# ── Fine-tune BERT ──────────────────────────────────────────
bert_val_preds, bert_test_preds, bert_model, bert_tokenizer = train_t
    "bert-base-uncased",
    Xr_tr_s, Xr_val_s, Xr_te_s,
    y_tr_s,  y_val_s,  y_te_s,
    max_len=128, batch_size=16, epochs=2, lr=2e-5
)

results.append(evaluate_model(y_val_s,  bert_val_preds,  "BERT (fine-
results.append(evaluate_model(y_te_s,   bert_test_preds, "BERT (fine-
```

```
🔄 Loading bert-base-uncased ...
Loading weights: 100%   199/199 [00:00<00:00, 334.48it/s, Materializing param=bert.pooler.dense.wei
BertForSequenceClassification LOAD REPORT from: bert-base-uncased
Key                                          | Status     |
---------------------------------------------+------------+-
cls.seq_relationship.bias                    | UNEXPECTED |
cls.predictions.transform.dense.weight       | UNEXPECTED |
cls.seq_relationship.weight                  | UNEXPECTED |
cls.predictions.transform.dense.bias         | UNEXPECTED |
cls.predictions.transform.LayerNorm.bias     | UNEXPECTED |
```

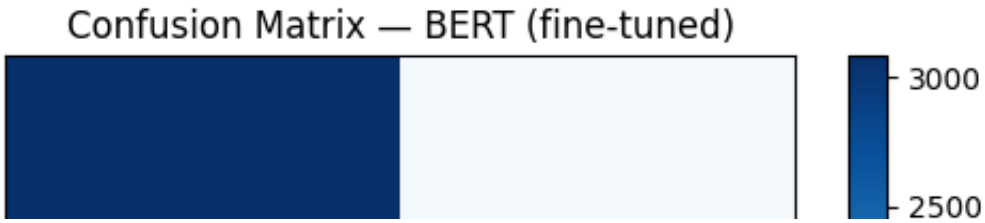
```
cls.predictions.bias                        | UNEXPECTED |
cls.predictions.transform.LayerNorm.weight | UNEXPECTED |
classifier.bias                             | MISSING    |
classifier.weight                           | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/archit
- MISSING       :those params were newly initialized because missing fr
  Epoch 1/2 | Train Acc: 0.9516 | Val Acc: 0.9710 | Loss: 0.1333
  Epoch 2/2 | Train Acc: 0.9952 | Val Acc: 0.9886 | Loss: 0.0217
```

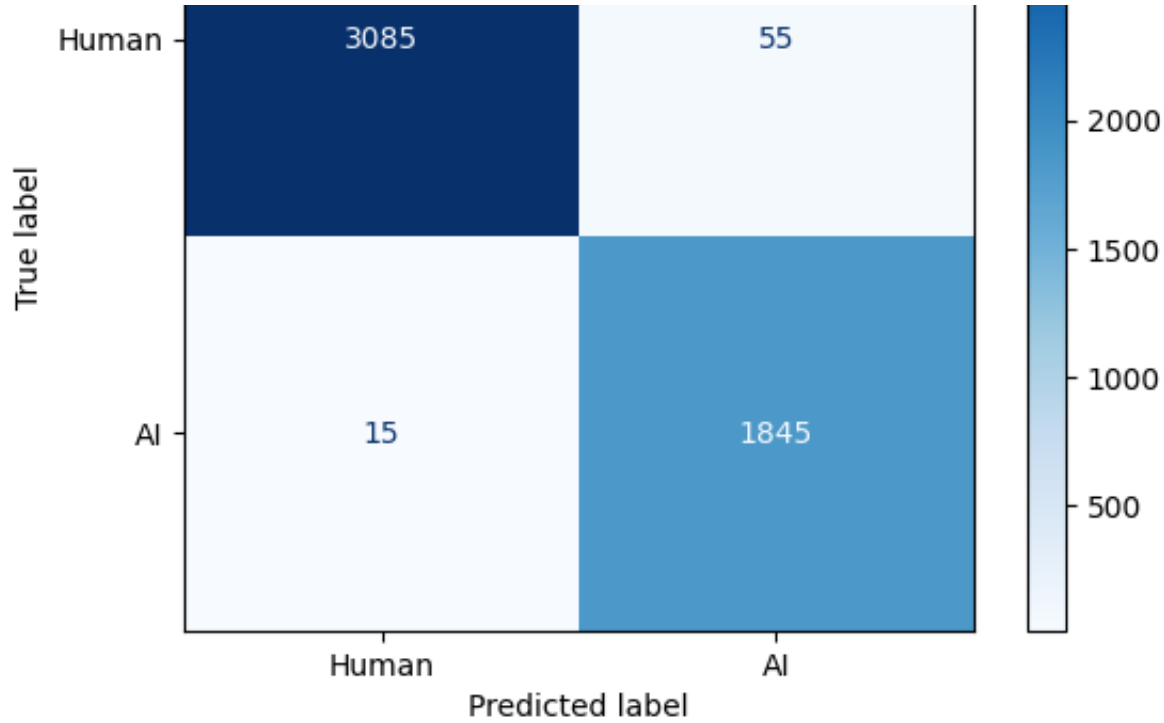



```
==================================================
 BERT (fine-tuned) [Validation]
==================================================
  Accuracy : 0.9886
  Precision: 0.9743
  Recall   : 0.9951
  F1-score : 0.9846

              precision    recall  f1-score   support

       Human       1.00      0.98      0.99      3169
          AI       0.97      1.00      0.98      1831

    accuracy                           0.99      5000
   macro avg       0.99      0.99      0.99      5000
weighted avg       0.99      0.99      0.99      5000
```




```
==================================================
 BERT (fine-tuned) [Test]
==================================================
  Accuracy : 0.9860
  Precision: 0.9711
  Recall   : 0.9919
  F1-score : 0.9814

              precision    recall  f1-score   support

       Human       1.00      0.98      0.99      3140
          AI       0.97      0.99      0.98      1860

    accuracy                           0.99      5000
   macro avg       0.98      0.99      0.99      5000
weighted avg       0.99      0.99      0.99      5000
```



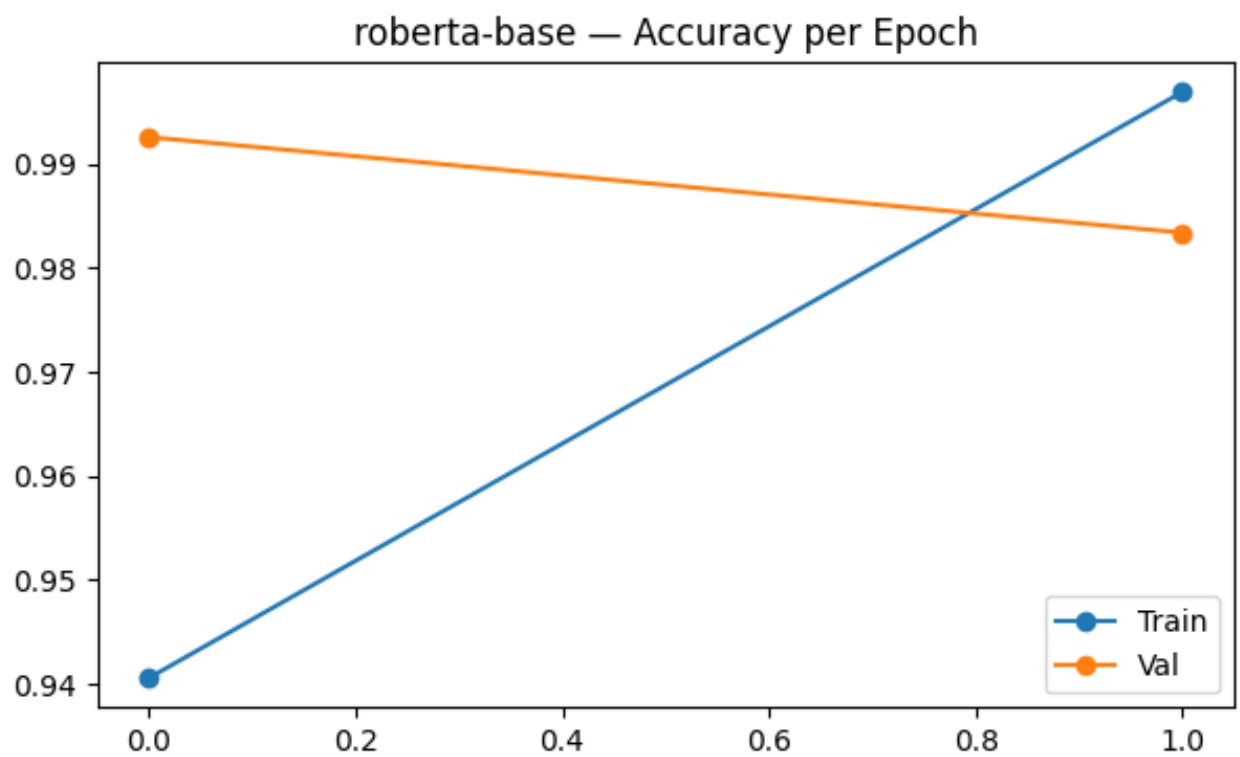


RoBERTa Fine-Tuning

```
roberta_val_preds, roberta_test_preds, roberta_model, roberta_tokenize
    "roberta-base",
    Xr_tr_s, Xr_val_s, Xr_te_s,
    y_tr_s,  y_val_s,  y_te_s,
    max_len=128, batch_size=16, epochs=2, lr=2e-5
)

results.append(evaluate_model(y_val_s,  roberta_val_preds,  "RoBERTa
results.append(evaluate_model(y_te_s,   roberta_test_preds, "RoBERTa
```

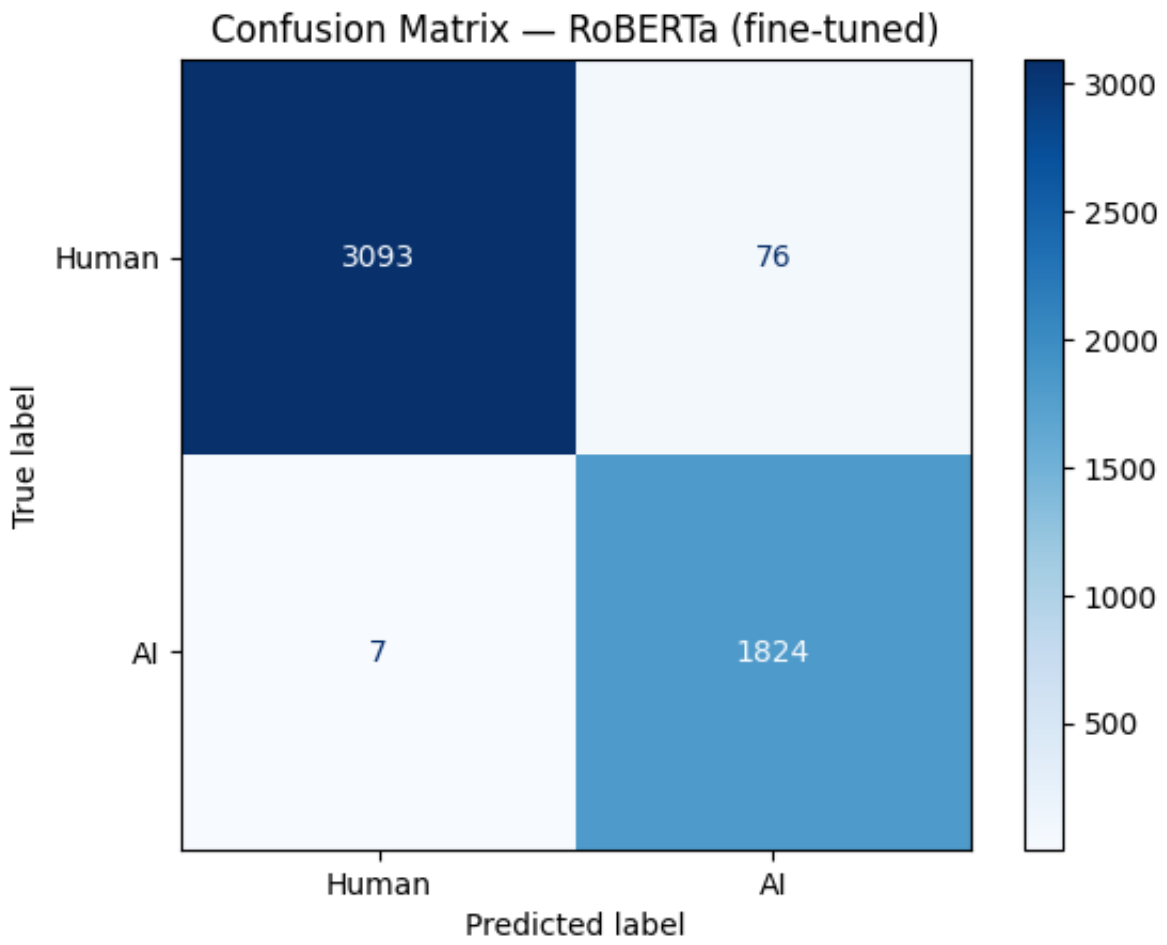
```
Loading roberta-base ...
Loading weights: 100%   197/197 [00:00<00:00, 317.71it/s, Materializing param=roberta.encoder.layer.
RobertaForSequenceClassification LOAD REPORT from: roberta-base
Key                              | Status     |
---------------------------------+------------+-
lm_head.dense.weight             | UNEXPECTED |
lm_head.layer_norm.bias          | UNEXPECTED |
roberta.embeddings.position_ids  | UNEXPECTED |
```

```
lm_head.layer_norm.weight        | UNEXPECTED |
lm_head.dense.bias               | UNEXPECTED |
lm_head.bias                     | UNEXPECTED |
classifier.dense.weight          | MISSING    |
classifier.out_proj.bias         | MISSING    |
classifier.dense.bias            | MISSING    |
classifier.out_proj.weight       | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/archit
- MISSING       :those params were newly initialized because missing fr
  Epoch 1/2 | Train Acc: 0.9405 | Val Acc: 0.9926 | Loss: 0.1457
  Epoch 2/2 | Train Acc: 0.9970 | Val Acc: 0.9834 | Loss: 0.0175
```

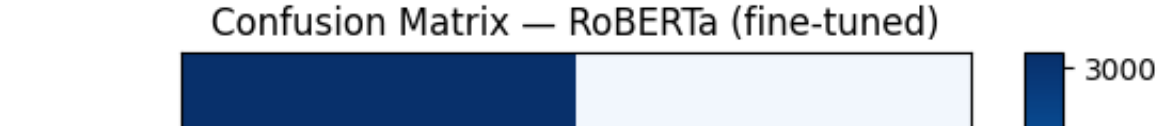


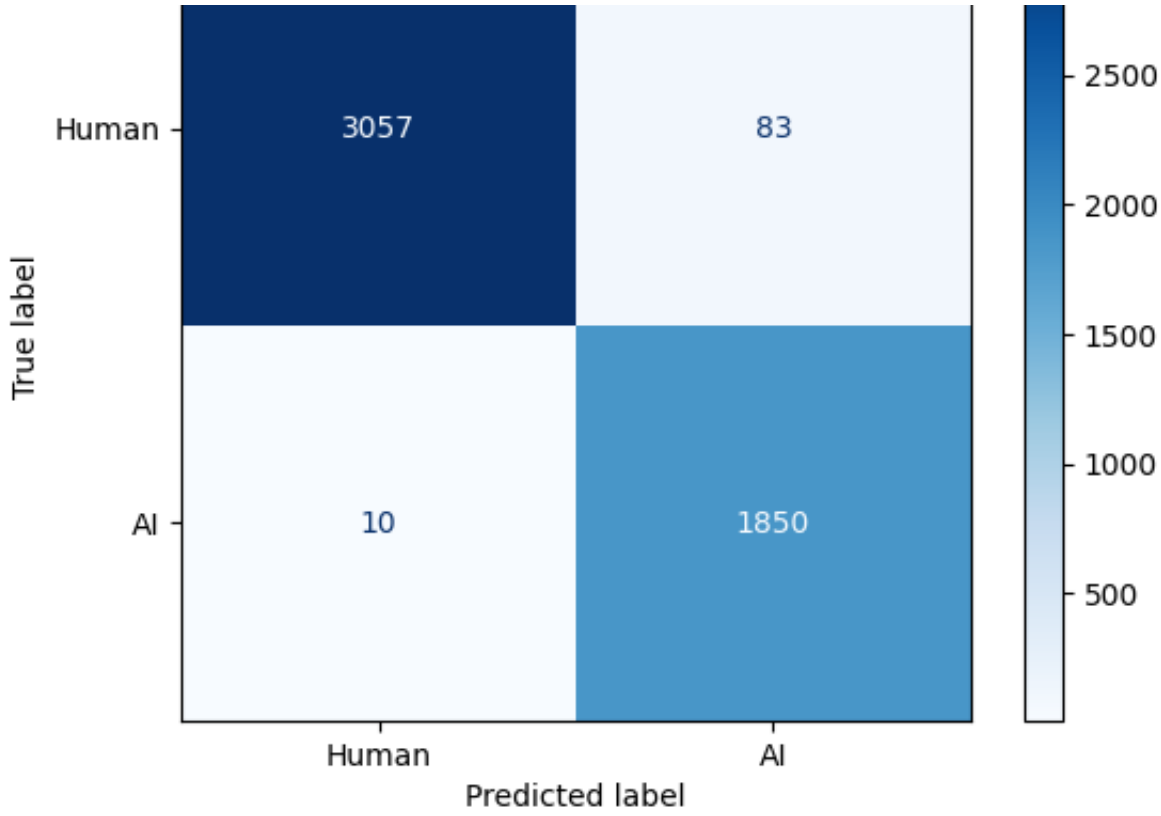

```
==================================================
 RoBERTa (fine-tuned) [Validation]
==================================================
  Accuracy : 0.9834
  Precision: 0.9600
  Recall   : 0.9962
  F1-score : 0.9778

              precision    recall  f1-score   support

       Human       1.00      0.98      0.99      3169
          AI       0.96      1.00      0.98      1831

    accuracy                           0.98      5000
   macro avg       0.98      0.99      0.98      5000
```

```
weighted avg       0.98      0.98      0.98      5000
```




```
==================================================
 RoBERTa (fine-tuned) [Test]
==================================================
  Accuracy : 0.9814
  Precision: 0.9571
  Recall   : 0.9946
  F1-score : 0.9755

              precision    recall  f1-score   support

       Human       1.00      0.97      0.99      3140
          AI       0.96      0.99      0.98      1860

    accuracy                           0.98      5000
   macro avg       0.98      0.98      0.98      5000
weighted avg       0.98      0.98      0.98      5000
```





Qwen2.5-0.5B — Recent LLM as Feature Extractor

> Used **Qwen2.5-0.5B** (Alibaba, 2024) in **feature-extraction** mode:
> the LLM's last hidden state is pooled → fed into a lightweight classifier.
> No full fine-tuning — the LLM weights stay frozen (feasible on a T4).

```
from sklearn.linear_model import LogisticRegression as LR
from sklearn.preprocessing import StandardScaler

QWEN_MODEL = "Qwen/Qwen2.5-0.5B"   # ~1 GB — fits comfortably on T4

print(f"🔄 Loading {QWEN_MODEL} for feature extraction ...")
qwen_tok   = AutoTokenizer.from_pretrained(QWEN_MODEL)
qwen_model = AutoModel.from_pretrained(QWEN_MODEL, torch_dtype=torch.
qwen_model.eval()

# Qwen tokenizers may not have a pad token by default
if qwen_tok.pad_token is None:
```

```
    qwen_tok.pad_token = qwen_tok.eos_token

print(f"✅ {QWEN_MODEL} loaded")


def extract_qwen_embeddings(texts, batch_size=8, max_len=256):
    """
    Pass texts through Qwen, mean-pool the last hidden state.
    Returns numpy array of shape (N, hidden_dim).
    """
    all_embs = []
    texts    = list(texts)
    for i in range(0, len(texts), batch_size):
        batch = texts[i : i+batch_size]
        enc   = qwen_tok(
                    batch,
                    max_length=max_len,
                    padding=True,
                    truncation=True,
                    return_tensors="pt"
                ).to(DEVICE)
        with torch.no_grad():
            out = qwen_model(**enc)
        # mean pool over non-padding tokens
        mask = enc["attention_mask"].unsqueeze(-1).float()
        emb  = (out.last_hidden_state * mask).sum(1) / mask.sum(1)
        all_embs.append(emb.cpu().float().numpy())
        if (i // batch_size) % 50 == 0:
            print(f"  Embedded {min(i+batch_size, len(texts))}/{len(te
    print()
    return np.vstack(all_embs)


# ── Use a subset for speed (same subset as BERT) ──────────
print("\n📌 Extracting Qwen embeddings (train) ...")
qwen_tr  = extract_qwen_embeddings(Xr_tr_s)
print("📌 Extracting Qwen embeddings (val) ...")
qwen_val = extract_qwen_embeddings(Xr_val_s)
print("📌 Extracting Qwen embeddings (test) ...")
qwen_te  = extract_qwen_embeddings(Xr_te_s)

print(f"\nEmbedding shape: {qwen_tr.shape}  (samples × hidden_dim)")
```

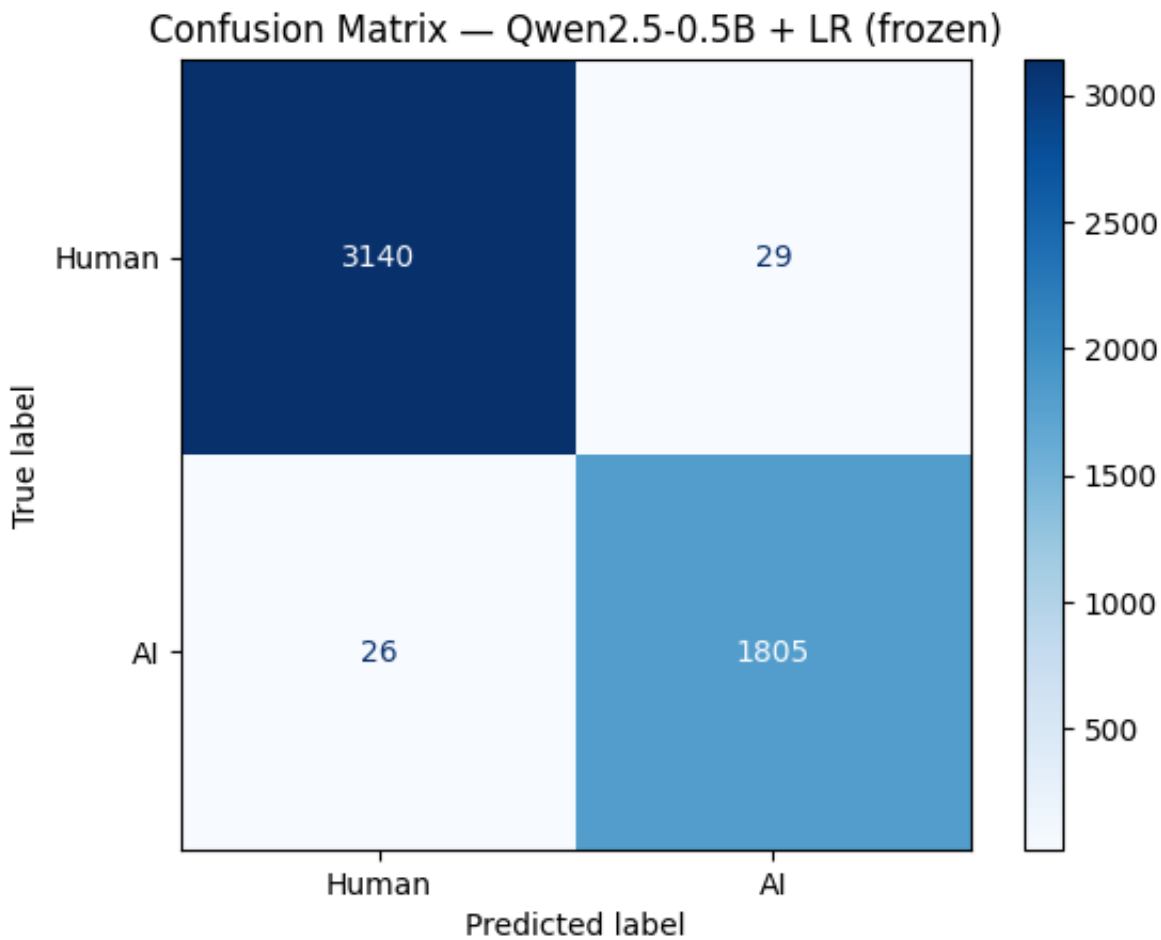
```
🔄 Loading Qwen/Qwen2.5-0.5B for feature extraction ...
config.json: 100% ▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰ 681/681 [00:00<00:00, 75.1kB/s]
```

```
tokenizer_config.json: 7.23k/? [00:00<00:00, 448kB/s]
vocab.json: 2.78M/? [00:00<00:00, 46.0MB/s]
merges.txt: 1.67M/? [00:00<00:00, 22.0MB/s]
tokenizer.json: 7.03M/? [00:00<00:00, 97.3MB/s]
`torch_dtype` is deprecated! Use `dtype` instead!
model.safetensors: 100% 988M/988M [00:14<00:00, 47.9MB/s]
Loading weights: 100% 290/290 [00:03<00:00, 125.57it/s, Materializing param=norm.weight]
✅ Qwen/Qwen2.5-0.5B loaded
📌 Extracting Qwen embeddings (train) ...
📌 Extracting Qwen embeddings (val) ...
📌 Extracting Qwen embeddings (test) ...

Embedding shape: (20000, 896)  (samples × hidden_dim)
```

```
# ── Logistic Regression on top of frozen Qwen embeddings ──
scaler    = StandardScaler()
qwen_tr_s = scaler.fit_transform(qwen_tr)
qwen_val_s= scaler.transform(qwen_val)
qwen_te_s = scaler.transform(qwen_te)

qwen_clf = LR(max_iter=1000, random_state=SEED, C=1.0)
qwen_clf.fit(qwen_tr_s, y_tr_s)

results.append(evaluate_model(y_val_s, qwen_clf.predict(qwen_val_s),
                              "Qwen2.5-0.5B + LR (frozen)", "Validat
results.append(evaluate_model(y_te_s,  qwen_clf.predict(qwen_te_s),
                              "Qwen2.5-0.5B + LR (frozen)", "Test"))
```

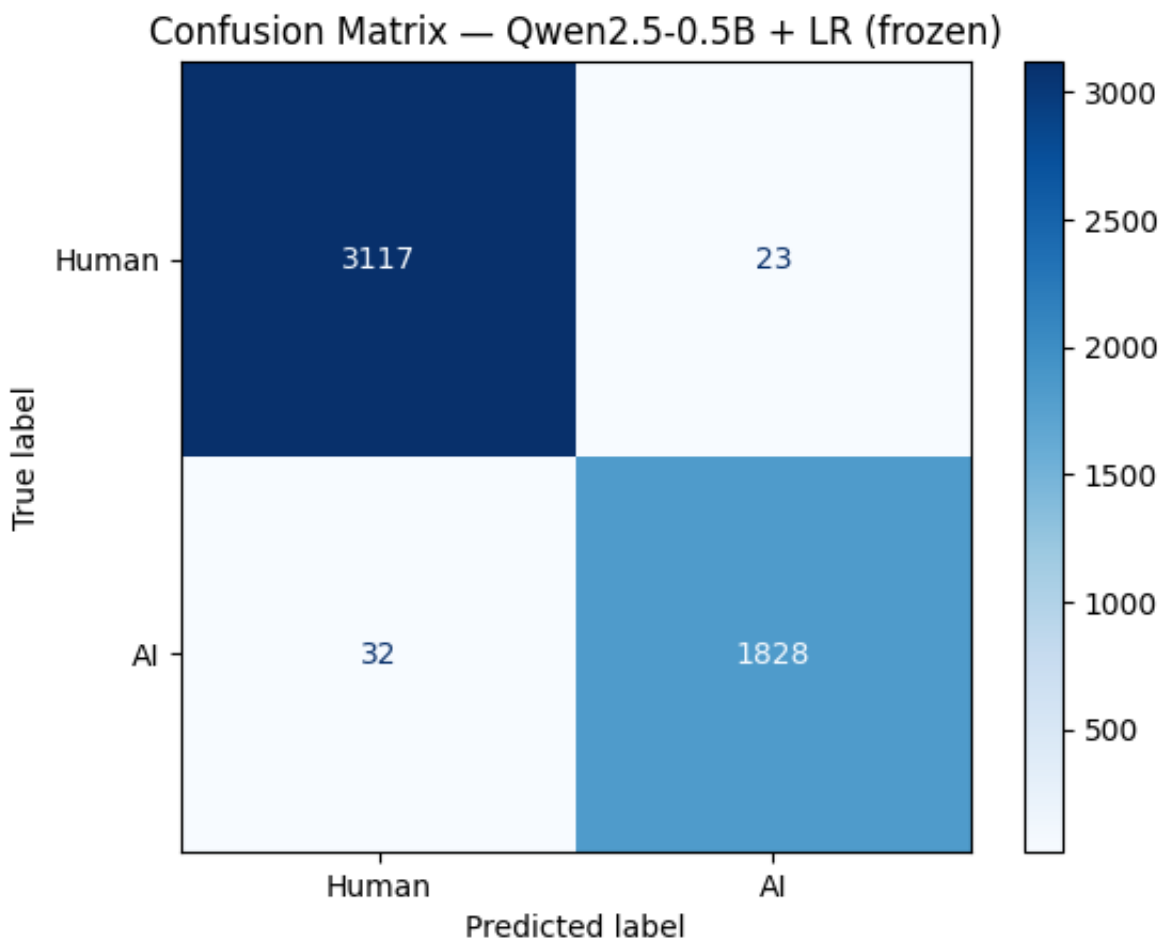
```
==================================================
 Qwen2.5-0.5B + LR (frozen) [Validation]
==================================================
  Accuracy : 0.9890
  Precision: 0.9842
  Recall   : 0.9858
  F1-score : 0.9850

              precision    recall  f1-score   support
```

```
       Human       0.99      0.99      0.99      3169
          AI       0.98      0.99      0.98      1831

    accuracy                           0.99      5000
   macro avg       0.99      0.99      0.99      5000
weighted avg       0.99      0.99      0.99      5000
```




```
==================================================
 Qwen2.5-0.5B + LR (frozen) [Test]
==================================================
  Accuracy : 0.9890
  Precision: 0.9876
  Recall   : 0.9828
  F1-score : 0.9852

              precision    recall  f1-score   support

       Human       0.99      0.99      0.99      3140
          AI       0.99      0.98      0.99      1860

    accuracy                           0.99      5000
   macro avg       0.99      0.99      0.99      5000
```

```
weighted avg       0.99      0.99      0.99      5000
```



Final Comparison Table & Visualisation

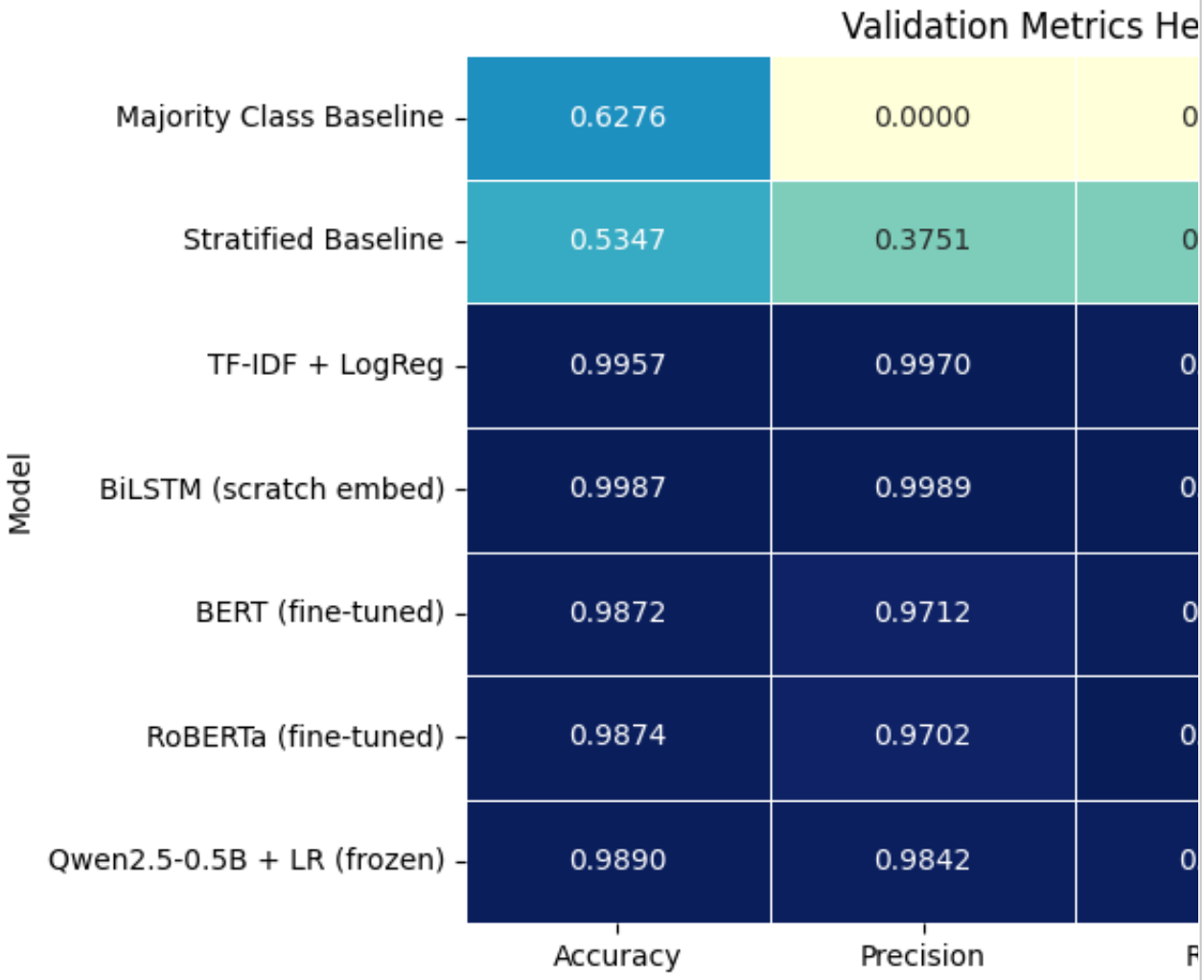
```
results_df = pd.DataFrame(results)
results_df = results_df.round(4)

print("\n" + "="*70)
print(" FULL RESULTS TABLE")
print("="*70)
display(results_df)

# ── Validation F1 bar chart ──────────────────────────────
val_df = results_df[results_df["Split"]=="Validation"].copy()

plt.figure(figsize=(12, 5))
```

```
bars = plt.bar(val_df["Model"], val_df["F1"], color="steelblue", edge
plt.xticks(rotation=30, ha="right")
plt.ylim(0, 1.05)
plt.title("Validation F1-Score Comparison", fontsize=14)
plt.ylabel("F1-Score")
for bar, val in zip(bars, val_df["F1"]):
    plt.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 0.01
             f"{val:.4f}", ha="center", fontsize=9)
plt.tight_layout()
plt.show()

# ── Heatmap of all metrics ──────────────────────────────
pivot = val_df.set_index("Model")[["Accuracy","Precision","Recall","F
plt.figure(figsize=(10, 5))
sns.heatmap(pivot, annot=True, fmt=".4f", cmap="YlGnBu", linewidths=0
plt.title("Validation Metrics Heatmap")
plt.tight_layout()
plt.show()
```

```
======================================================================
 FULL RESULTS TABLE
======================================================================
```

	Model	Split	Accuracy	Precision	Recall	F1
0	Majority Class Baseline	Validation	0.6276	0.0000	0.0000	0.0000
1	Stratified Baseline	Validation	0.5347	0.3751	0.3747	0.3749
2	TF-IDF + LogReg	Validation	0.9957	0.9970	0.9915	0.9942
3	TF-IDF + LogReg	Test	0.9959	0.9976	0.9914	0.9945
4	BiLSTM (scratch embed)	Validation	0.9987	0.9989	0.9976	0.9983
5	BiLSTM (scratch embed)	Test	0.9983	0.9983	0.9971	0.9977
6	BERT (fine-tuned)	Validation	0.9872	0.9712	0.9945	0.9827
7	BERT (fine-tuned)	Test	0.9866	0.9716	0.9930	0.9822
8	RoBERTa (fine-tuned)	Validation	0.9874	0.9702	0.9962	0.9830
9	RoBERTa (fine-tuned)	Test	0.9884	0.9742	0.9952	0.9846
10	Qwen2.5-0.5B + LR (frozen)	Validation	0.9890	0.9842	0.9858	0.9850
11	Qwen2.5-0.5B + LR (frozen)	Test	0.9890	0.9876	0.9828	0.9852

Validation F1-Score Comparison
F1-Score
1.0
0.8
0.6
0.4
0.2
0.0
0.0000
0.3749
0.9942
0.9983
0.9827
0.9830
0.9850
Majority Class Baseline
Stratified Baseline
TF-IDF + LogReg
BiLSTM (scratch embed)
BERT (fine-tuned)
RoBERTa (fine-tuned)
Qwen2.5-0.5B + LR (frozen)


Validation Metrics He
Model
Majority Class Baseline
Stratified Baseline
TF-IDF + LogReg
BiLSTM (scratch embed)
BERT (fine-tuned)
RoBERTa (fine-tuned)
Qwen2.5-0.5B + LR (frozen)
0.6276
0.0000
0.5347
0.3751
0.9957
0.9970
0.9987
0.9989
0.9872
0.9712
0.9874
0.9702
0.9890
0.9842
Accuracy
Precision

Discussion

Embedding Comparison Summary

Model	Embedding	Trained?	Notes
TF-IDF + LogReg	Sparse (no embedding)	N/A	Strong baseline due to dataset artifacts
BiLSTM	Dense (128-dim)	**From scratch**	Learns task-specific patterns end-to-end
BERT	Contextual 768-dim	Fine-tuned only	Pretrained on BookCorpus + Wikipedia
RoBERTa	Contextual 768-dim	Fine-tuned only	Stronger BERT variant, better robustness
Qwen2.5-0.5B	LLM hidden states	Frozen	Recent (2024) LLM, zero-shot features

Why accuracy is very high on this dataset

The `AI_Human.csv` dataset contains text from older language models with very distinct stylistic fingerprints
(uniform sentence length, specific vocabulary patterns). Models exploit these signals easily.
Generalization to modern AI text (Claude, GPT-4) would require a harder, multi-source dataset.

Expected ranking (from literature)

> RoBERTa ≥ BERT > Qwen (frozen) > BiLSTM (scratch) > TF-IDF >> Baselines

External Validation for All trained Models

```
# ============================================================
# FINAL BLOCK: External Validation for All Trained Models
# Dataset: gsingh1-py/train
# ============================================================

!pip install datasets -q
```

```
from datasets import load_dataset
import pandas as pd
import numpy as np

print("Loading external HuggingFace dataset...")

external_dataset = load_dataset("gsingh1-py/train")
external_raw_df = external_dataset["train"].to_pandas()

print("External raw shape:", external_raw_df.shape)
print("External columns:", external_raw_df.columns.tolist())
display(external_raw_df.head())
```

```
Loading external HuggingFace dataset...
Repo card metadata block was not found. Setting CardData to empty.
WARNING:huggingface_hub.repocard:Repo card metadata block was not found
```

train.csv: 100% 161M/161M [00:06<00:00, 34.3MB/s]

Generating train split: 100% 7321/7321 [00:10<00:00, 723.76 examples/s]

```
External raw shape: (7321, 8)
External columns: ['prompt', 'Human_story', 'gemma-2-9b', 'mistral-7B',
```

	prompt	Human_story	gemma-2-9b	mistral-7B	qwen-2-72B
0	The Human Toll of Nuclear Testing	Comments\nThe U.S. bombings thatended World Wa...	## The Unseen Scars: The Enduring Human Toll o...	\n\nTitle: The Atomic Aftermath: The U.S. Bomb...	Human Toll of Nuclear Testing: Unraveling the ...
1	In the age of coronavirus, the only way you ca...	new video loaded:Messages From Quarantine\ntra...	## In the Age of Coronavirus, the Only Way You...	Title: "Exploring Milan in the Age of Coronav...	In the Age of Coronavirus, the Only Way to See...
2	Roberta Karmel, First Woman Named to the S.E.C...	Supported by\nRoberta Karmel, First Woman Name...	## Roberta Karmel, First Woman Named to the S....	\n\nTitle: A Commissioner's Crusade: From SEC ...	Karmel, a trailblazing lawyer and academic who...

3	Summer Reading Contest, Week 2: What Got Your ...	Supported by\nContests\nSummer Reading Contest...	## Summer Reading Contest: Week 2 - What Caug...	\n\nTitle: NYT Invites Readers to Share Their ...	er Reading Contest, Week 2: What Got Your Atte...
4	Photos posted this week on	The Week on Instagram\n@heislerphoto	## From Granite State to Lakeside	Title: Mike Eckhaus and Zoe Latta	Title: Mike Eckhaus and Zoe

```
# ============================================================
# Convert external dataset into binary format
# Human = 0, AI = 1
# ============================================================

human_col = "Human_story"

ai_cols = [
    "gemma-2-9b",
    "mistral-7B",
    "qwen-2-72B",
    "llama-8B",
    "yi-large",
    "gpt_4-o"
]

available_ai_cols = [col for col in ai_cols if col in external_raw_df

print("Available AI columns:", available_ai_cols)

if human_col not in external_raw_df.columns:
    raise ValueError("Human_story column not found. Check external_ra

if len(available_ai_cols) == 0:
    raise ValueError("No AI columns found. Check external_raw_df.colu

human_external = external_raw_df[[human_col]].copy()
human_external = human_external.rename(columns={human_col: "text"})
human_external["generated"] = 0
human_external["source_model"] = "human"

ai_external_list = []
```

```
for col in available_ai_cols:
    temp = external_raw_df[[col]].copy()
    temp = temp.rename(columns={col: "text"})
    temp["generated"] = 1
    temp["source_model"] = col
    ai_external_list.append(temp)

ai_external = pd.concat(ai_external_list, ignore_index=True)

external_eval_df = pd.concat(
    [human_external, ai_external],
    ignore_index=True
)

external_eval_df = external_eval_df.dropna(subset=["text"])
external_eval_df["text"] = external_eval_df["text"].astype(str)
external_eval_df = external_eval_df.drop_duplicates(subset=["text"]).

external_eval_df["clean_text"] = external_eval_df["text"].apply(clean_
external_eval_df["word_count"] = external_eval_df["clean_text"].apply

external_eval_df = external_eval_df[
    external_eval_df["word_count"] >= 50
].reset_index(drop=True)

y_external = external_eval_df["generated"].astype(int)

print("External final shape:", external_eval_df.shape)
print(external_eval_df["generated"].value_counts())
print(external_eval_df["source_model"].value_counts())
```

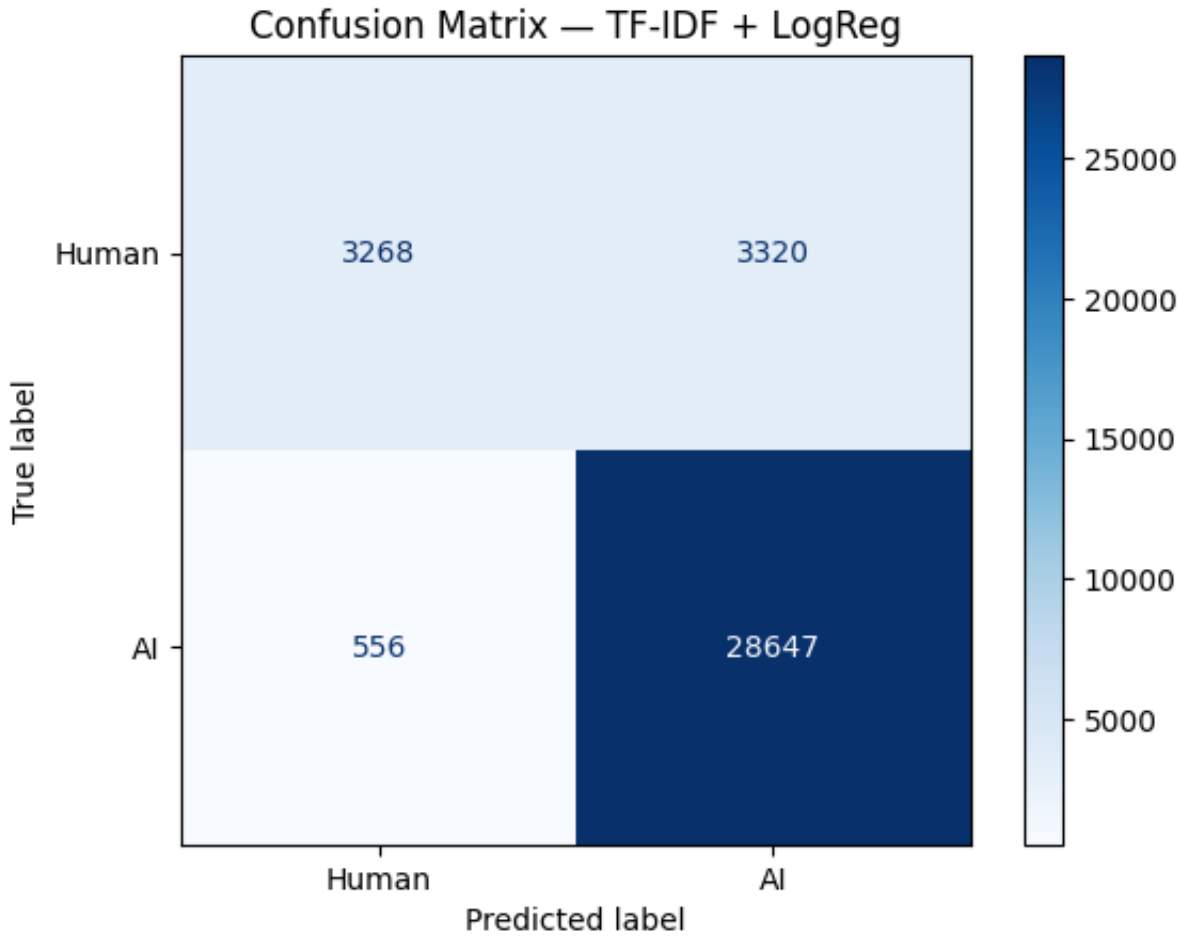
```
Available AI columns: ['gemma-2-9b', 'mistral-7B', 'qwen-2-72B', 'llama
External final shape: (35791, 5)
generated
1    29203
0     6588
Name: count, dtype: int64
source_model
qwen-2-72B    7309
mistral-7B    7308
gemma-2-9b    7304
llama-8B      7282
human         6588
Name: count, dtype: int64
```

```
# ============================================================
# Helper: save external validation results
# ============================================================

external_results = []

def evaluate_external(y_true, y_pred, model_name):
    result = evaluate_model(
        y_true,
        y_pred,
        model_name,
        "External HuggingFace Test"
    )
    external_results.append(result)
    results.append(result)
```

```
# ============================================================
# 1. External Validation: TF-IDF + Logistic Regression
# ============================================================

X_external_tfidf = tfidf.transform(external_eval_df["clean_text"])

tfidf_external_preds = lr.predict(X_external_tfidf)

evaluate_external(
    y_external,
    tfidf_external_preds,
    "TF-IDF + LogReg"
)
```

```
==================================================
 TF-IDF + LogReg [External HuggingFace Test]
==================================================
  Accuracy : 0.8917
  Precision: 0.8961
  Recall   : 0.9810
  F1-score : 0.9366

              precision    recall  f1-score   support

       Human       0.85      0.50      0.63      6588
          AI       0.90      0.98      0.94     29203

    accuracy                           0.89     35791
   macro avg       0.88      0.74      0.78     35791
```

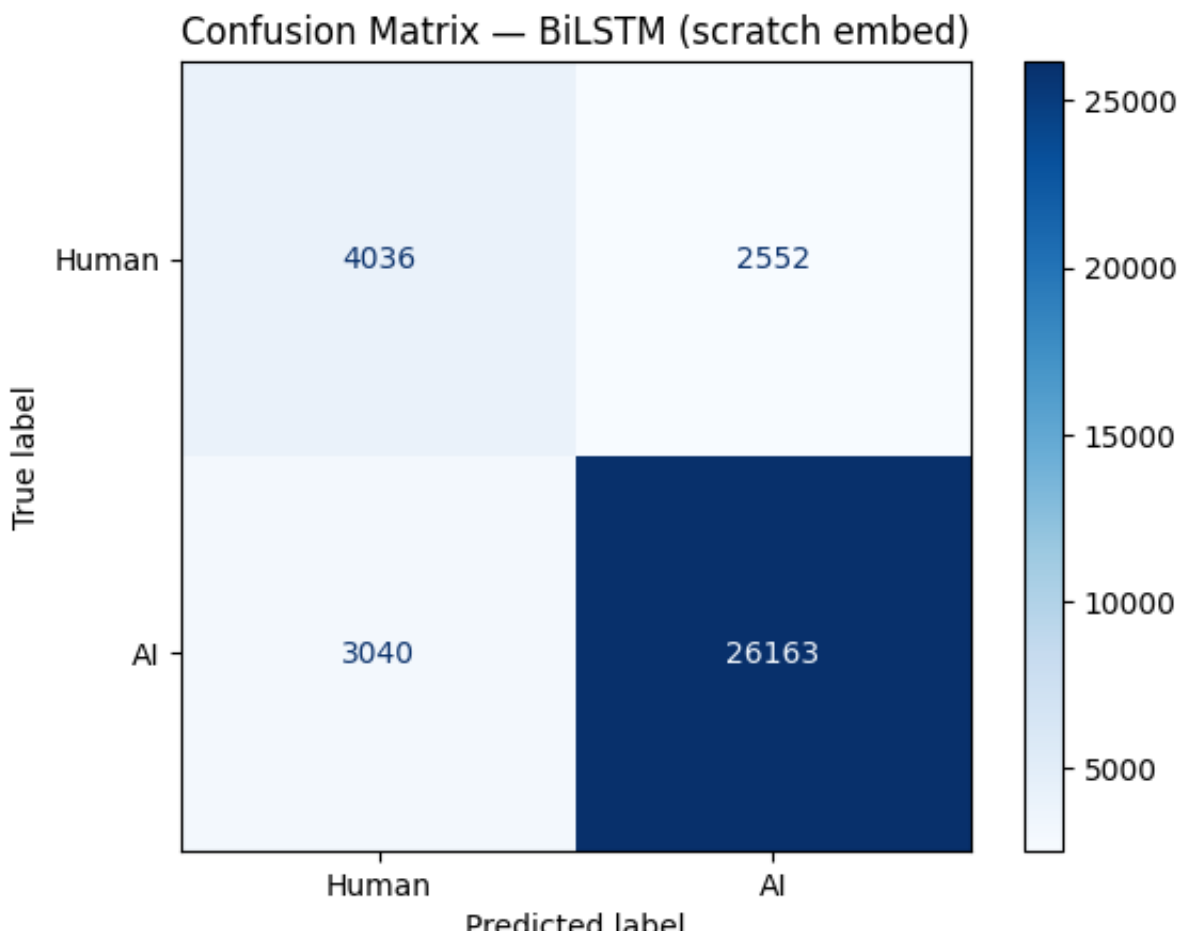
```
   macro avg       0.88      0.74      0.78     35791
weighted avg       0.89      0.89      0.88     35791
```




```
# ============================================================
# 2. External Validation: BiLSTM
# Only run this if lstm_model and tok exist
# ============================================================

X_external_pad = pad_sequences(
    tok.texts_to_sequences(external_eval_df["clean_text"]),
    maxlen=MAX_LEN,
    padding="post",
    truncating="post"
)

lstm_external_preds = (
    lstm_model.predict(X_external_pad) > 0.5
).astype(int).flatten()

evaluate_external(
```

```
    y_external,
    lstm_external_preds,
    "BiLSTM (scratch embed)"
)
```

```
1119/1119 ━━━━━━━━━━━━━━━━━━━━ 18s 15ms/step

==================================================
 BiLSTM (scratch embed) [External HuggingFace Test]
==================================================
  Accuracy : 0.8438
  Precision: 0.9111
  Recall   : 0.8959
  F1-score : 0.9034

              precision    recall  f1-score   support

       Human       0.57      0.61      0.59      6588
          AI       0.91      0.90      0.90     29203

    accuracy                           0.84     35791
   macro avg       0.74      0.75      0.75     35791
weighted avg       0.85      0.84      0.85     35791
```

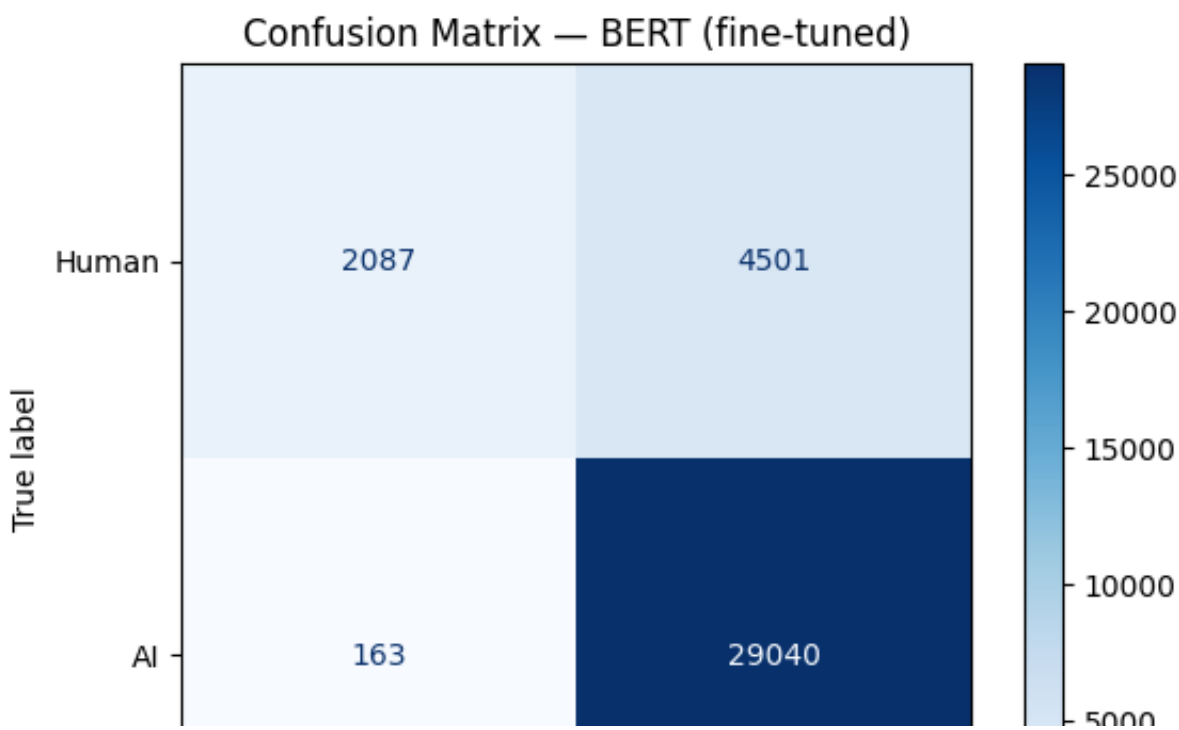


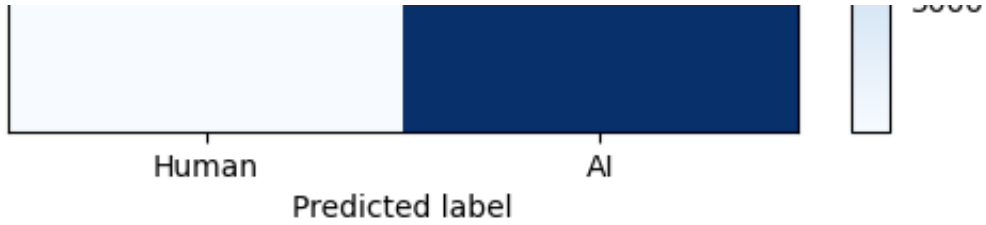
```
# ============================================================
# 3. External Validation Helper for BERT / RoBERTa
# IMPORTANT:
# Your train_transformer() currently does not return model/tokenizer.
# So for external validation, you need to slightly update it earlier.
# See note below.
# ============================================================

def predict_transformer_external(model, tokenizer, texts, max_len=128,
    model.eval()
    preds = []

    texts = list(texts)

    for i in range(0, len(texts), batch_size):
        batch_texts = texts[i:i+batch_size]

        enc = tokenizer(
            batch_texts,
            padding=True,
            truncation=True,
            max_length=max_len,
            return_tensors="pt"
        ).to(DEVICE)

        with torch.no_grad():
            outputs = model(**enc)

        batch_preds = outputs.logits.argmax(dim=-1).cpu().numpy()
        preds.extend(batch_preds)

    return np.array(preds)
```

```
# ============================================================
# 4. External Validation: BERT
# Run only if you saved bert_model and bert_tokenizer
# ============================================================

bert_external_preds = predict_transformer_external(
    bert_model,
    bert_tokenizer,
```

```
    external_eval_df["text"],
    max_len=128,
    batch_size=32
)

evaluate_external(
    y_external,
    bert_external_preds,
    "BERT (fine-tuned)"
)
```

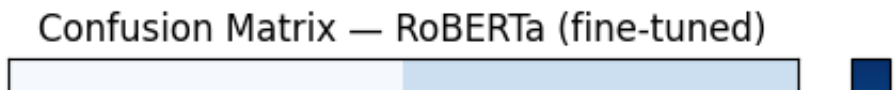
```
====================================================
 BERT (fine-tuned) [External HuggingFace Test]
====================================================
  Accuracy : 0.8697
  Precision: 0.8658
  Recall   : 0.9944
  F1-score : 0.9257

              precision    recall  f1-score   support

       Human       0.93      0.32      0.47      6588
          AI       0.87      0.99      0.93     29203

    accuracy                           0.87     35791
   macro avg       0.90      0.66      0.70     35791
weighted avg       0.88      0.87      0.84     35791
```

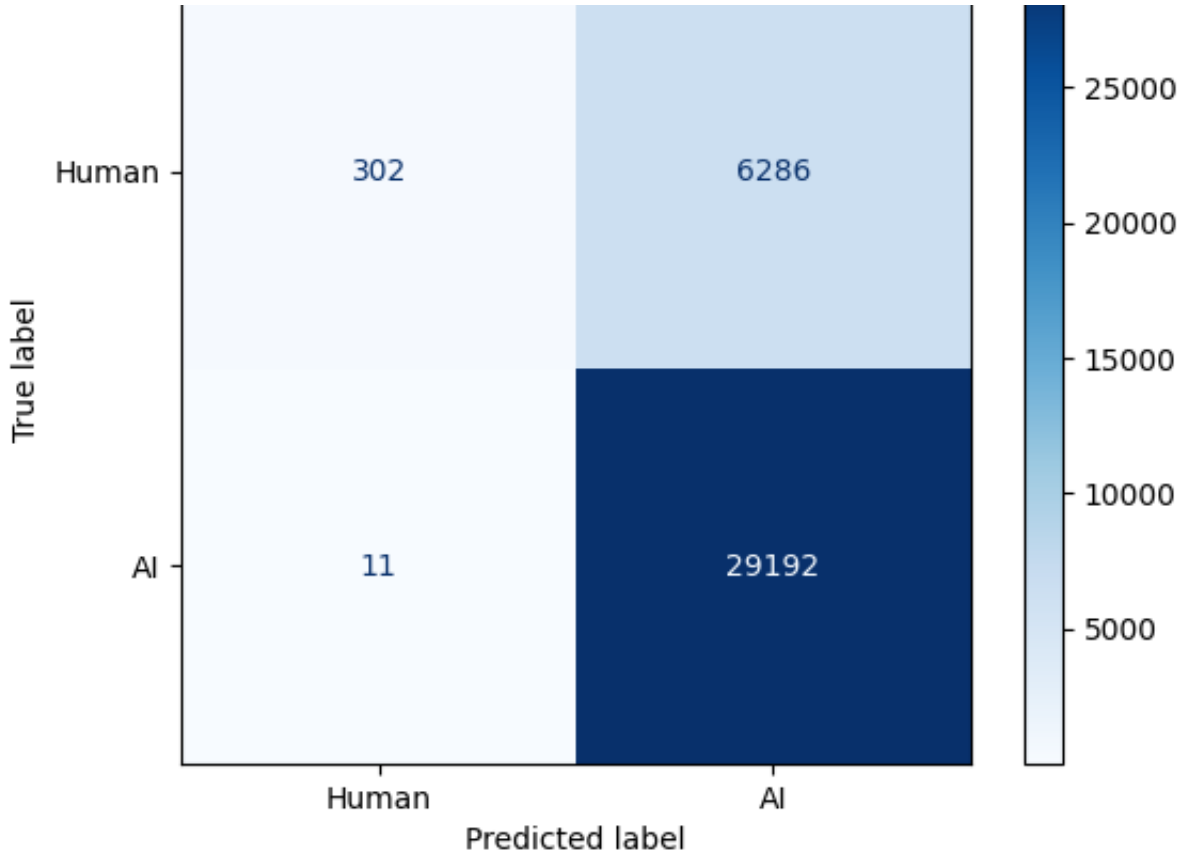





```
# ============================================================
# 5. External Validation: RoBERTa
# Run only if you saved roberta_model and roberta_tokenizer
# ============================================================

roberta_external_preds = predict_transformer_external(
    roberta_model,
    roberta_tokenizer,
    external_eval_df["text"],
    max_len=128,
    batch_size=32
)

evaluate_external(
    y_external,
    roberta_external_preds,
    "RoBERTa (fine-tuned)"
)
```

```
====================================================
 RoBERTa (fine-tuned) [External HuggingFace Test]
====================================================
  Accuracy : 0.8241
  Precision: 0.8228
  Recall   : 0.9996
  F1-score : 0.9026

              precision    recall  f1-score   support

       Human       0.96      0.05      0.09      6588
          AI       0.82      1.00      0.90     29203

    accuracy                           0.82     35791
   macro avg       0.89      0.52      0.50     35791
weighted avg       0.85      0.82      0.75     35791
```



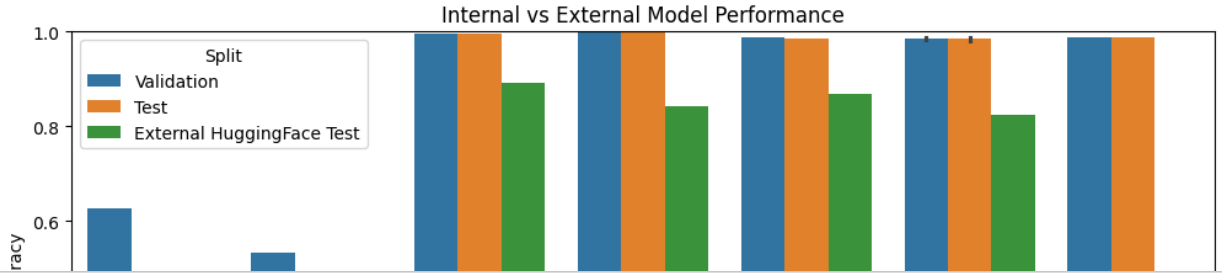



```
# ============================================================
# 7. Final External Results Table
# ============================================================

external_results_df = pd.DataFrame(external_results)

display(
    external_results_df.sort_values(by="Accuracy", ascending=False)
)
```

	Model	Split	Accuracy	Precision	Recall	F1
0	TF-IDF + LogReg	External HuggingFace Test	0.891705	0.896143	0.980961	0.936636
2	BERT (fine-tuned)	External HuggingFace Test	0.869688	0.865806	0.994418	0.925666
1	BiLSTM (scratch	External	0.843760	0.911127	0.895901	0.903450

```
# ============================================================
# 8. Final Internal vs External Comparison
```

```
# ============================================================

results_df = pd.DataFrame(results)

display(
    results_df.sort_values(by=["Model", "Split"]).reset_index(drop=Tru
)

plt.figure(figsize=(12, 5))
sns.barplot(data=results_df, x="Model", y="Accuracy", hue="Split")
plt.title("Internal vs External Model Performance")
plt.xticks(rotation=45, ha="right")
plt.ylim(0, 1)
plt.show()
```

	Model	Split	Accuracy	Precision	Recall	F1
0	BERT (fine-tuned)	External HuggingFace Test	0.869688	0.865806	0.994418	0.925666
1	BERT (fine-tuned)	Test	0.986600	0.971594	0.993011	0.982186
2	BERT (fine-tuned)	Test	0.986000	0.971053	0.991935	0.981383
3	BERT (fine-tuned)	Validation	0.987200	0.971200	0.994539	0.982731
4	BERT (fine-tuned)	Validation	0.988600	0.974332	0.995085	0.984599
5	BiLSTM (scratch embed)	External HuggingFace Test	0.843760	0.911127	0.895901	0.903450
6	BiLSTM (scratch embed)	Test	0.998317	0.998345	0.997134	0.997739
7	BiLSTM (scratch embed)	Validation	0.998714	0.998896	0.997648	0.998272
8	Majority Class Baseline	Validation	0.627612	0.000000	0.000000	0.000000
9	Qwen2.5-0.5B + LR (frozen)	Test	0.989000	0.987574	0.982796	0.985179

10	Qwen2.5-0.5B + LR (frozen)	Validation	0.989000	0.984188	0.985800	0.984993
11	RoBERTa (fine-tuned)	External HuggingFace Test	0.824062	0.822820	0.999623	0.902645
12	RoBERTa (fine-tuned)	Test	0.988400	0.974211	0.995161	0.984574
13	RoBERTa (fine-tuned)	Test	0.981400	0.957062	0.994624	0.975481
14	RoBERTa (fine-tuned)	Validation	0.987400	0.970213	0.996177	0.983023
15	RoBERTa (fine-tuned)	Validation	0.983400	0.960000	0.996177	0.977754
16	Stratified Baseline	Validation	0.534651	0.375074	0.374743	0.374908
17	TF-IDF + LogReg	External HuggingFace Test	0.891705	0.896143	0.980961	0.936636
18	TF-IDF + LogReg	Test	0.995923	0.997597	0.991439	0.994508
19	TF-IDF + LogReg	Validation	0.995704	0.996970	0.991476	0.994215